Floating Bridge Overhead Contact System
Expansion Joints
Guidance



Course Content Begins at 4.3.1

4.3.1 I-90 Floating Bridge

Floating Bridge Overhead Contact System Expansion Joints Guidance

Historical Technical Documentation | Professional Portfolio Sample

This document represents my work developing technical guidance in support of rail electrical workforce training and development.

Created within a safety-critical transit environment, the material illustrates my approach to organizing complex engineering and maintenance information into structured, accessible technical documentation that supports instructor-led learning and workforce competency development.

The document reflects a historical stage in the evolution of a living maintenance standard. Subsequent operational developments, engineering decisions, and ongoing revisions may have superseded portions of its contents.

It is presented as evidence of my technical writing, instructional design, and documentation capabilities.

Historical Context: This document is not a current or operational reference. The underlying work remains subject to ongoing development and revision, and the material presented here has been overtaken by subsequent events and developments.

CONTENTS

1.0 ABOUT THIS STANDARD....2
1.1 Purpose and Objectives....2
1.2 Scope....2
1.2.1 Scope Exclusions....3
1.3 Intended Audience....3
1.4 Overview....3
2.0 DEFINITIONS....4
3.0 PROCEDURAL REQUIREMENTS....18
3.1 Roles and Responsibilities....18
3.2 Qualified Persons....19
3.2.1 Work Performed by Others....20
3.3 Operational Records....20
3.3.1 Single Line Diagrams (SLDs)....20
3.3.2 Configuration Files....20
3.4 Incident Logging....20
3.5 Single-Point Data Consolidation and Trending Required....21
3.6 Priority Rating (Based on APTA-RT-FS-S-006-03, Rev. 2 Section 8, SMP's, and EAMS)....21
4.0 SYSTEM OVERVIEW....22
4.1 OCS Technical Specifications....22
4.2 Feeding Arrangement....24
4.3 Unique OCS Features....24
4.3.1 I-90 Floating Bridge....24
4.3.2 Electromagnetic Interference (EMI) Section from University of Washington Station (UWS) to U-District Station (UDS)....24
5.0 COMPONENTS AND ASSEMBLIES....25
5.1 Insulated Overlaps....25
5.2 Uninsulated Overlaps....25
5.3 Tensioning Devices....25
5.3.1 Balance Weight Assemblies (BWA)....25
5.3.2 Constant Tension Spiral Spring Tensorex C+....26
5.3.3 Variable Tension Linear Spring....26
5.3.4 Fixed Termination....26

5.4 Section Insulators....26
5.5 I-90 Floating Bridge OCR Expansion Joints....26
5.6 I-90 Floating Bridge Catenary Monitoring System....27
5.7 Midpoint anchors....27
5.8 OCR....27
5.9 Crossovers....27
5.10 Supports....27
5.10.1 Poles....27
5.10.2 Drop Posts....28
5.10.3 Portals....28
5.10.4 Cantilevers....28
5.10.5 Cross Spans....28
5.10.6 Hangers....28
5.10.7 Grounding....28
5.11 Surge arresters....29
5.12 Maintenance and Storage Facilities....29
6.0 CONDITION ASSESSMENT....29
6.1 Condition Rating Methodology....29
6.2 Condition Assessment Measure TERM Scale....30
6.3 Condition Assessment Reporting....33
7.0 FUNCTIONAL TESTING....33
7.1 Cables....33
7.1.1 Visual and mechanical inspections (per APTA-RT-FS-S-006-03 Rev 2 Section 6.2)....33
7.1.2 Electrical Tests....34
7.1.3 Test Values....34
7.2 For additional testing, reference sample OCS Test Procedures included and attached with this OCS Maintenance Standard in Appendices 10.4 and 10.5....34
8.0 PREVENTATIVE MAINTENANCE TASK TABLE....35
8.1 OCS Maintenance Task Table....35
8.2 OCR Maintenance Task Table....63
9.0 REFERENCES....73
9.1 IEEE Std 1791-2019 IEEE Recommended Practice for Terminology Used for Direct Current Electric Transit Overhead Contact Systems....73

9.2 APTA-RT-FS-S-006-03-Rev 2 – 2017 Traction Electrification Distribution System Inspection, Maintenance and Testing....73
9.3 IGA 2019 KCC Ordinance....73
9.4 Sound Transit Requirements Manual (ST RM) Rev 1 – DEC 2024....73

10.0 APPENDICES....73

10.1 Appendix A – Asset Hierarchy....75
10.2 Appendix B – Torque Specification Tables....83

10.2.1 US standard Fasteners - Heat-Treated Steel Cadmium or Zinc Plated....83
10.2.2 US Standard Fasteners – Silicon Bronze....83
10.2.3 US Standard Fasteners – Aluminum Alloy....84
10.2.4 US Standard Fasteners – Stainless Steel....84

10.3 Appendix C – OCS Maintenance Standard RACI (Responsible, Accountable, Consulted, Informed) Matrix....86
10.4 Appendix D – OCS Acceptance Measurements (per ST Standard Specs 34 23 69 and ST RM)....87
10.5 Appendix E – Reference Sample OCS Test Procedures attached with this standard....91

10.5.1 N830 E750-REC-04216 SUB 342369-014.002 1.03A.1.b OCS Acceptance Measurements....91
10.5.2 N830 E750-REC-04215 SUB 342369-008.002 1.03A.1.b OCS Visual Inspection 91
10.5.3 N830 E750-REC-04274 SUB 342369-012.002 1.03A.1.b OCS Electrical & Mechanical Clearance Test Procedure....91
10.5.4 N830 E750-REC-04276 SUB 342369-007.002 1.03A.1.b OCS Continuity Test Procedure....91
10.5.5 N830 E750-REC-04277 SUB 342369-004.002 1.03A.1.b OCS Hi-Pot Dielectric....91
10.5.6 N830 E750-REC-04275 SUB 342369-006.002 1.03A.1.b OCS Earth Ground Resistance....91
10.5.7 N830 E750-REC-06390 SUB 019500-048.003 1.05A SIT 301-302 OCS Live Wire 91

10.6 Appendix F – N830-E750 NG EL MW & CW Temperature/Tensions Charts....92
10.7 Appendix G – L800 LLE MW & CW Temperature/Tensions Charts....95
10.8 Appendix H – F200 FWLE MW and CW Temperature/Tensions Charts....96
10.9 Appendix I – U830 University Link MW and CW Temperature/Tensions Charts....98
U830 Contract IFC Dwg No. L10-JO702, Sheet 547 for CW....98
U830 Contract IFC Dwg No. L10-JO703, Sheet 548 for MW....99
10.10 Appendix J – Figure 1 – 350 MCM Contact Wire Wear from SouthWear....100
10.11 Appendix K – Figure 2 – Tensorex C+ Spiral Spring Tensioning Device (by Mosdorfer) for Auto-Tensioned (AT) OCS....101

LIST OF TABLES

Table 2-1 Definitions and Abbreviations
Table 4-1 Climatic Conditions
Table 6-1 Condition Assessment Matrix
Table 6-2 Condition Rating Matrix Example
Table 6-3 Condition Rating Index Weighting
Table 8-1 OCS Maintenance Task Table
Table 8-2 OCR Maintenance Task Table

LIST OF FIGURES/ CHARTS

10.6 Appendix F – N830-E750 Northgate East Link Ext MW & CW Temperature/Tensions Charts
10.7 Appendix G – L800 Lynnwood Link Extension MW & CW Temperature/Tensions Charts
10.8 Appendix H – F200 Federal Way Link Extension MW and CW Temperature/Tensions Charts
10.9 Appendix I – U830 University Link Extension MW and CW Temperature/Tensions Charts
10.10 Appendix J – Figure 1 – 350 MCM Contact Wire Wear.....100
10.11 Appendix K – Figure 2 – Tensorex C+ Spiral Spring Tensioning Device (by Mosdorfer) for Auto-Tensioned (AT) OCS.

Executive Summary

Sound Transit Maintenance Standards provide guideline for testing, inspecting, and performing predictive and preventative maintenance on Sound Transit assets. This Maintenance Standard establishes roles and responsibilities of persons involved in the development and execution of maintenance programs, describes how the system's equipment and assemblies' function, and establishes requirements for condition reporting, scheduled maintenance, and record keeping ensuring safe and reliable operation of the system.

1.0 ABOUT THIS STANDARD

This document establishes a standard for predictive and preventative maintenance of Overhead Contact System (OCS) equipment and assemblies. It provides a baseline for predictive and preventative maintenance at Sound Transit and facilitates financial forecasting and life-cycle cost analysis.

Adherence to this Standard ensures that appropriate maintenance tasks are performed and documented based on the condition of the asset, and it provides a means for identifying and predicting failures that affect system reliability, availability, and safety.

This Standard will be revised as additional facilities, services, equipment, and assemblies are added to the system and as lessons are learned.

1.1 Purpose and Objectives

This Standard establishes minimum requirements for those responsible for overseeing, facilitating, assessing, inspecting, and maintaining the OCS. This Standard provides a baseline by which the Sound Transit assets can be managed, evaluated, and maintained.

This Standard intends to meet the following objectives:

i. Establish inspection and maintenance criteria to ensure OCS equipment is safe, operational, and reliable.

ii. Allow Sound Transit to predict asset end of life and inform their financial forecasting.

iii. Allow Sound Transit to review and improve upon maintenance practices.

1.2 Scope

This document includes minimum requirements for the preventative maintenance of OCS equipment. The scope of this Standard includes the following Systems.

i. 1 Line (Lynnwood – Angle Lake, north to south link extensions).

ii. 2 Line (Seattle – Downtown Redmond, west to east link extensions).

   a. At the time of writing, portions of the Link 2 Line remain under construction, including the corridor travelling across the Lake Washington I-90 Floating Bridge, which connects Downtown Seattle to Bellevue and Redmond.

iii. OMF Central (OMF-C)

iv. OMF East (OMF-E)

1.2.1 Scope Exclusions

This Standard does not provide detailed preventative maintenance task orders, and it excludes requirements for corrective maintenance and repairs.

This does not include other systems, such as Link T Line or Sounder. A separate document, or revision to this document, will include Link T Line and planned system expansions such as lines to:

i. West Seattle
ii. Ballard
iii. Everett
iv. Tacoma Dome Link Extension

1.3 Intended Audience

This Standard is intended for use by Sound Transit, King County Metro Maintenance of Way (MOW) and applicable Contractors. It provides necessary information to:

i. SMEs and consultants developing detailed preventative maintenance procedures.

ii. EAMS program managers and administrators involved in the development of the EAMS system.

iii. Asset managers.

iv. Sound Transit Systems Engineering team.

v. Those performing systems tests.

vi. Those seeking to improve Sound Transit's maintenance program.

1.4 Overview

Section 1 provides information about this standard and includes an overview the purpose and objectives for these Maintenance Standards. It introduces the document's intended audience, establishes its scope, and introduces its organization and content.

Section 2 provides definitions related to the OCS.

Section 3 describes procedural requirements. It establishes the roles and responsibilities of those involved in the maintenance of Sound Transit assets, including requirements for appointing qualified persons to perform maintenance tasks. It also provides requirements for record keeping and how to properly respond when non-conformance or failure is encountered.

Section 4 and Section 5 provides a high-level overview of OCS. It describes the overall system as well as other systems that impact or rely upon the OCS. Section 5 describes parts, components, and assemblies within the system. These descriptions are intended to provide a knowledgeable audience with a basic understanding of ST's OCS so that they can conceptualize preventative maintenance task from a programmatic perspective.

Section 6 establishes requirements for condition assessments. Sound Transit assets are required to be assessed at standard intervals. The results of the condition assessment influence the frequency and types of preventative maintenance inspections and tasks required for the asset and facilitate asset management and financial forecasting.

Section 7 provides requirements for functional testing of the OCS.

Section 8 provides the minimum requirements for the type and frequency of inspection and maintenance tasks that must be performed on OCS equipment and assemblies.

Section 9 Appendices.

Section 10 provides sample forms and guidance for visual condition assessment inspections, functional testing procedures, and other documentation to assist users of this standard in implementing the requirements of this Maintenance Standard.

2.0 DEFINITIONS

These definitions provide uniform terminology and definitions for technical terms used in design of the Overhead Contact System. These terms will form the basis for assembly descriptions used in material allocation control. The list does not include standard transit and engineering terms but defines terms for which industry standard definitions have not been universally adopted, especially those related to the OCS.

Table Definitions- 1 Definitions and Abbreviations

Term	Definition
Acceptance Measurements	Final acceptance records of the installed contact and messenger wires heights and staggers at the overhead contact system (OCS) supports and registrations, and at midspan. Also known as *as-builts*.
Along-Track Movement (ATM)	The motion of the catenary induced by counterweights or tension equipment due to thermal expansion or contraction of the conductors.
Anchor	• Anchor Bolt: A bolt inserted into a concrete foundation to form an attachment for a pole or support member/bracket.

Term	Definition
	• Anchor Plate: o A plate typically embedded at the end of an anchor rod foundation used with down guys. o An attachment to the face of a structure or foundation for securing the overhead contact system (OCS).
Anchor Bracket Assembly (CR-AB)	CR-AB is a steel structure used to support the OCR transition from OCS wires to rigid conductor bar.
Anchor Clamp	Part of the Anchor Bracket Assembly. This provides strong support and attachment of transition bar into the track.
Arrangement (OCS)	A group of structures in a specific configuration providing the wiring of a typical short section of OCS (e.g., midpoint anchor, overlap, crossover).
Assembly	A configuration of components used repetitively in the OCS and identified in combination for ease of reference.
Auto Tensioned (AT)	Catenary conductors terminated with counterweights, including balance weight assemblies and spring tensioners such as Tensorex C+ devices. These tensioning devices maintain constant tension over a specified range of conductor temperatures by compensating for conductor thermal elongation and contraction.
Balance Weight Assembly (BWA)	The tensioning equipment with weights and devices that are installed at each end of a tension length of an auto-tensioned overhead contact system (OCS) to maintain constant wire tensions over a known range of wire extension. The BWAs play a critical role in maintaining a safe and healthy overhead line and must be monitored and maintained frequently.
Bar Insulator	Part of the Anchor Bracket Assembly. This assembly consists of insulated bar and standoffs that insulate the weldment structures to the OCR bar. The sliding hanger clamp accommodates the nature of the track that crosses the anchor bracket at an angle laterally.
Bond	An electrical connection between metal hardware that minimizes voltage difference. • Impedance Bond: An inductive device bridging an insulated rail joint used to allow the passage of dc traction return current.

Term	Definition
	• Rail Bond: an electrical connection between adjacent lengths of rail. • Structure Bond. An electrical connection between a structure and the rebar cage of the foundation and/or a ground rod, or a grounding circuit of a civil structure.
Bracket	A connection to a pole or support member by means of which it is attached to the OCS assembly attachment.
Bridle	A span wire or guy wire between structural supports used primarily to provide vertical support to a cross-span wire attached approximately midway along the bridle.
Cable Cleat	A mechanical device that secures cable to provide support and prevent movement during normal operation and especially during fault conditions like short circuits.
Cable Support Assembly	Part of the Jumper Assembly. This consists of an insulated cable clamp and are installed on column steel structure of CR-EXP. One is located on top of the steel tower and the other along the back column steel structure. Both serve the purpose of holding the 2kV rated insulated cable.
Cantilever	The support frame and registration assembly supporting the catenary from a pole, attached to pole or structure via hinged fittings.
Catenary	The combination of conductors, hangers, and in-span hardware of the overhead contact system used in LRT operation, not including supports. • Dictionary meaning - the curve assumed by a perfectly flexible cord of uniform density and cross-section hanging freely from two fixed points. • Simple Catenary - comprising of a contact wire supported from a messenger wire by hangers. • Compound Catenary - comprised of a contact wire, an auxiliary wire, and a messenger wire. • Single Wire - comprising of a contact wire only.
Catenary Monitoring System	The I-90 floating bridge is more vulnerable to stray current corrosion than other utility structures. To ensure prompt detection of broken OCS wires, which can generate significant stray current Siemens Sicat Catenary Monitoring System (CMS) was installed along I-90 floating bridge. It continuously monitors the OCS for large and significant changes in the tensile

Term	Definition
	force between the contact wire and messenger wires. The event status will be detected quickly and transmitted to the link control center.
Chafing	Damage or wear caused by excessive rubbing.
Clamp	• Parallel Groove Clamp - a piece of hardware used to clamp two parallel wires together. • Pipe Clamp - a piece of hardware used to attach various types of components to a pipe. • Strain Clamp - a piece of hardware used for dead-ending a wire or conductor under high tension. • Suspension Clamp - a piece of hardware used to support a tensioned conductor or cable in a hanging arrangement, the greater part of the applied load being due to gravity.
Component	An item of hardware as commonly supplied complete by manufacturers.
Compound Catenary	See “Catenary.”
Compression Lug	Cable terminations, used for bonding or feeder connections.
Contact Wire	The wire on which the pantograph of a light rail vehicle (LRV) is designed to make contact for current collection.
Contact Wire Bridge	Contact Wire Bridge Assembly which may also be referred as Cross Contact is used at crossovers. It is a rigid bar, up to 5 feet long, fixed closely above and attached to the in running contact wire forming a slot for a second contact wire to pass through, thereby preventing differential uplift of crossing contact wires.
Contact Wire Clamp	Part of the Expansion Joint Assembly. Single CW clamp used for transition bars. Other CW clamps such as at (registration) steady arms and pull offs.
Contact Wire Height	The height of the underside of the contact wire above top of centerline of rail when not uplifted by the pantograph. • Minimum contact wire height - the minimum allowable contact wire height, usually at mid-span or under bridges, which can be used taking into account vehicle clearances envelope, vehicle bounce and track tolerances, catenary temperature effects and electrical clearances. May also take future track rising into account if so required. • Maximum contact wire height - the maximum allowable contact wire

Term	Definition
	height within the operating range of the vehicle pantograph.
Creep	The ongoing permanent stretch of wires under tension for long periods of time.
Cross-Level	The plane through the tops of the rails of a track in the transverse direction
Cross span	A wire that crosses the track or roadway at approximately contact wire level, more specifically a single wire fastened to poles on either side of the route and supporting a contact or trolley wire.
Crossover	A special rail assembly that allows the LRV to cross from one track to another.
Current Collector	The carbon strips on the top of a pantograph that engages the contact wire.
Curve	• Curve Span: A cross span in a curve with one or more double curve hangers or a curve segment. • Inside Curve: An arrangement for an OCS where the poles are on the inside of the curve. • Outside Curve: Poles are on the outside of the curve.
Dead-end or Fixed End (Termination)	A conductor under tension that is terminated without a tensioning device.
Disconnect Switch	A switching device that isolates one OCS segment from another or a switching device that isolates the traction power substation feeder breaker from the OCS.
Double Curve Hanger	An overhead assembly used in cross-spans to support trolley wires in large radius curves. Each hanger can provide up to 7 degrees deviation in the trolley wires.
Double Insulation	Two levels of insulation. Insulation typically provided by two physically separated insulators.
Electrical Clearance	• Passing (dynamic) - the dimensional mechanical clearance between live parts of either the vehicle (such as current collector) or OCS and grounded parts of the fixed structures, or during the passing of the vehicle. • Static - the dimensional mechanical clearance between live parts of the OCS and grounded parts of fixed structures, under non-passing conditions.
Employee in Charge (EIC)	The designated KCM employee, with a current KCM Light Rail EIC certification, which oversees a work group. The EIC shall be assisted by flaggers when necessary.

Term	Definition
Enterprise Asset Management System (EAMS)	Maintenance Tracking Software. Trapeze software is used within Sound Transit to manage maintenance, inventory and repair work for operational assets and equipment.
Expansion Joint Assembly (CR-EXP)	CR-EXP is used to provide continuous power to the light rail vehicle travelling the floating bridge's expansion joints.
Feeders	Conductors that supply power to or augment the power-carrying capacity of the conductors in an overhead contact system.
Fittings	Small components used in assembly of cantilever pipes and catenaries.
Fixed Terminated (FT) Equipment	OCS with dead-ended conductors.
Gauge	• Load gauge - the envelope around the track within which all loaded track vehicles are specified to remain while static or in motion. • Track gauge - the distance between the inside running edges of the track. • Electrical umbrella - the electrical clearance envelope for the pantograph in the swayed position over its operating height range. • Pole gauge - the thickness of steel in a tubular pole.
Ground Wire	The conductor installed for the purpose of providing electrical continuity between the supporting structure of the overhead contact system and the common return or grounding system.
Guy	A steadying or positioning wire. • Down or back guy - a wire attached high to a pole and coming down at an angle to an anchor in the ground. • Head or span - a wire between two points but not anchored to the ground. • Sidewalk down guy - a down guy that is diverted to a vertical position for anchorage to the ground, by use of a horizontal strut braced against the pole at an elevated level. Used where the normal down guy anchorage would occur on a sidewalk or other pathway.
Half-tension	A sectionalizing point in an overhead contact system (OCS) formed by two

Term	Definition
Air-break	parallel contact wires, each at half tension, and cut into a single contact wire over 30.48 m (100 ft) to 60.96 m (200 ft). Half-tension air-breaks do not require additional contact wire anchor poles, thus making them a practical substitute for section insulators.
Half-tension length	In auto-tensioned (AT) equipment, the segment of overhead contact system (OCS) between a midpoint anchor and tensioning device (such as a balance weight assembly (BWA) or Tensorex C+), or between a fixed termination and tensioning device (BWA or Tensorex C+).
Hanger	A vertical fitting by means of which the contact wire is suspended from the messenger wire at regular intervals (typically every 15-30 feet).
Head-Span	An installation of wires that crosses the track or roadway at approximately contact wire level, more specifically a double or triple wire fastened to poles on either side of the route and supporting a catenary, contact or trolley wire.
Head or Span Guy	A wire between two points but not anchored to the ground.
Horns	The curved or angled downwards ends of the pantograph head which ensure movement of out-of-running wires onto the current collector.
Impedance Bond	See “Bond.”
In-Running (IR) Catenary	The catenary in multiple catenary sections (such as overlaps) providing power to the pantograph.
Insert	A mechanical screw socket-type fixing for direct embedment into concrete usually placed before concreting.
Insulated Support Assembly	Part of the Expansion Join Assembly. Consists of two sets of heavy-duty standoff insulators that have mounting plate provision on both sides. The wider mounting plate always attaches to the main column structure.
Insulator	Any body or substance provided and designed for the purpose of surrounding or supporting a conductor to restrict the flow of electricity to a desired path. • Disc - a bell-shaped insulator of glass or porcelain used singly or in strings. • Johnny Ball or Porcelain Strain Insulator - a type of strain insulator used primarily for guy or span wire insulation. • Section - see “Section Insulator.”

Term	Definition
	• Standoff - a solid core insulator with bending strength. • Strain - an insulator or a string of disc insulators used in line with a tensioned conductor. • Strut - a standoff insulator used in compression members of cantilevers. • Suspension - an insulator or string of discs that are suspended in vertical position. • Synthetic - an insulator made from fiberglass plastic or epoxy resin.
Jumper	Generally, an electrical connection in the overhead contact system; a short conductor installed to provide electrical continuity. • Continuity or full current: A jumper capable of carrying full line current from one catenary to another longitudinally at tensioning overlaps and track turnouts. • Equalizing: A light internal jumper in the OCS, connecting the messenger to the contact wire, generally once in each span.
Jumper Assembly (CR-XJ)	CR-XJ is used to transfer electricity from one side of the expansion joint to the other because the parallel overlapping OCR bars are separated by 18 inches.
Light Rail Vehicle (LRV)	A single bodied or articulated electrically powered rail passenger car, capable of operating on private right-of-way, city streets, or reserved center strips in freeways or city streets as required by route alignment.
Link Control Center (LCC)	The primary location for controlling, monitoring, and dispatching the Link Light Rail system.
Live	• Live load (structural) - a load or force that is temporary in nature such as wind, ice, the dynamic uplift force of a passing pantograph. • Live wire (electrical) - a conductor carrying a voltage for power supply.
LPSCAT	• Low-Profile, Simple Catenary Auto-Tensioned (LPSCAT).
LPSCFT	• Low Profile, Simple Catenary Fixed Terminated (LPSCFT).
Lug	• Crimped - An attachment to the end of a wire for an electrical connection made using a gripping or crimping tool. • Terminal - A crimped or a soldered piece to terminate a wire for electrical

Term	Definition
	connection.
Maintenance of Way (MOW)	Includes the mainline, maintenance yard and pocket tracks, switches, and structures (tunnels, floating slabs, bridges, aerial structures, culverts) and the upkeep of the entire Link System.
Maximum Operating Wind Speed	The maximum wind speed, in which vehicles will continue to be operated at their normal speed, which for design is typically 55MPH. The wind speed selected for pantograph security calculations is shown in Table System Overview-2 Climatic Conditions.
Messenger Wire	The uppermost conductor in a catenary system. The conductor that hangs in the approximate shape of a catenary from which the contact wire is suspended by means of hangers.
Midpoint Anchor	The point approximately midway between two tensioning devices (BWAs or Tensorex C+) of an Auto-Tensioned OCS wire run where the catenary is anchored against along-track movement.
Normally Closed (NC)	Describes the position of a disconnect or sectionalizing switch when its normal operating configuration is to be closed and connecting two OCS segments or connecting the substation DC feeder breaker to the OCS.
OCR Support Column	Part of the Expansion Joint Assembly, this sub-assembly holds the roller support and termination assembly. This must be installed to match the called-out OCS contact/messenger wire direction. This item determines the approximate nominal contact wire (CW) height.
Offset	• The deviation of the static contact wire from the normal centerline of the track or roadway. • The dimension of the centerline of the pole from the centerline of the track or roadway.
Overhead Conductor Rail (OCR)	Consists of an extruded aluminum profile that accommodates the copper contact wire, with a great cross section for the current that allows operative OCS voltages.
Overhead Contact System (OCS)	That part of the traction power system – comprising of the catenary (or single contact wire), supply and return feeders, catenary supports, foundations, counterweights or tensioning devices and other equipment and assemblies – that deliver electric power from the traction substation to rail vehicles.
Overlap	That portion of the overhead contact system where the contact and messenger

Term	Definition
	wires of two adjoining sections overlap before terminating at opposite ends.
Pantograph	A current collection device fitted on top of a LRV, hinged to vary in height as it engages the contact wire.
Pantograph Head	The uppermost part of the pantograph which is fitted with the current collector.
Pantograph Security	The analysis of the dimensional relationship between pantograph and contact wire that allows for vehicle and pantograph sway, track tolerances, super-elevation, erection tolerances and wind factor on wires and poles. Also includes temperature effects on AT and FT equipment.
Pantograph Sway	Maximum lateral displacement of the pantograph induced by vehicle roll and lateral shock loads.
Parallel Feeder	An along-track aerial bare or insulated cable mounted on the OCS poles, or an insulated cable installed in a trough or duct, which provides electrical power reinforcement to the OCS by means of feeder jumpers at regular intervals.
Pocket Track	An extra (usually center) track within a double track system or a portion of track within a terminal on which an LRV may be stored for a period.
Pole (also OCS Pole)	A vertical pole structure with or without a guy, used to support the overhead contact system.
Portal Assembly (All Types)	Used across the I-90 Floating bridge. Consists of two columns and a member across the top connecting the two, with the corners reinforced with braces. Each column has a hinge on the baseplate, which is used to mitigate the effects of along track forces. Unlike traditional OCS poles, steel portals are electrically isolated from the bridge structure at the hinge base to address stray current control considerations. An aerial grounding cable has been designed to be attached to each portal structure so that all portals are still grounded to either bank of the lake.
Power Feed Block	Part of the Jumper Assembly and Anchor Bracket Assembly. This block consists of a bimetallic plate and provides connection between the insulated cable and the OCR bar.
Profile (Overhead)	The vertical alignment of the OCS conductors relative to the top of rail.
Pulley	A pulley block used in messenger support and catenary termination systems.
Qualified	A person who is familiar and knowledgeable in the construction and operation

Term	Definition
Electrical Employee (QEE)	of the electric power generation, transmission, and distribution equipment involved, and such lines and/or equipment that concerns their position and who is fully aware of potential hazards related to performance of the work, and who has passed a journey status examination for the particular branch of the electrical trades with which they are connected.
Radial Load	The horizontal load on a structure applied by conductors due to deviation by registration and/or track curvature.
Rail Electrical Worker (REW)	A high voltage qualified electrical employee assigned to the Traction Power group.
Rake	A lean preset of an OCS pole from vertical, before loading, such that when loaded the pole does not lean forward from vertical.
Registration	Lateral support of conductors to maintain a fixed location.
Right of Way (ROW)	The area and property that extend 10 feet from the nearest rail and/or property that is dedicated to Light Rail mainline operations.
Riser	A vertical cable supplying power upward from the feeder system to the overhead conductors.
Roller Support	Part of the Expansion Joint Assembly. Holds the OCR ramp bar.
Running Edge	The inside faces of the rail heads of a track.
Sag	The difference between the average heights of the conductor at adjacent supports and its height at the lowest point in the span.
Sagging	The act of installing messengers and other conductors to correct tension by measuring conductor sag.
SCADA	Supervisory control and data acquisition (SCADA) system.
SCAT	Simple catenary auto-tensioned (SCAT) overhead contact system.
SCFT	Simple catenary fixed termination (SCFT) overhead contact system.
Section Insulator	A device for dividing a contact wire (and messenger wire) into electrical sections while maintaining mechanical continuity and a continuous path for the current collectors.
Sectionalizing Switch	A disconnect switch installed at sectionalizing points that are normally closed for train operations. Used for isolating overhead line sections.
Sectioning or	The division of an electrical distribution system or network into electrical

Term	Definition
Sectionalizing	sections.
Sleeve	• Pipe sleeve: A short length of smaller diameter pipe fitted into a larger diameter pipe to reduce the internal diameter of the larger pipe. • Sleeve foundation: A circular concrete tube placed in a drilled hole to provide a foundation. • Reinforcing sleeve or ground sleeve, also referred to as a “dog collar”: A steel sleeve placed around a steel pole at ground line to protect against corrosion damage. • Repair sleeve: A compressed copper or aluminum sleeve around a conductor to repair local electrical damage.
Soffit	The underside of a tunnel roof or bridge deck.
Span	The horizontal length of trolley or contact wire between two adjacent support points (not necessarily the distance or difference in stationing between the support structures).
Splice	A fitting for joining two conductors together and capable of carrying full current capacity of the conductors through the joint. May be a bolted or compression fitting.
Splice Plate	Part of the Expansion Joint Assembly. Lines up and connects OCR bars.
Stagger	The (normally alternating) offset of the contact wire from the superelevated track centerline by registration at each support, to cause the contact wire to sweep over the pantograph head during vehicle operation.
Steady Arm or Registration Arm	The lateral restrainer on the contact wire at a structure.
Stiffening Weldment	Part of the Anchor Bracket Assembly. This weldment provides additional strength to the vertical support and keeps them square.
Stray Current and Corrosion Control Remote Monitoring System (SCCCRMS)	The Remote Monitoring shall provide for measurement of track-to-earth isolation-to-earth isolation by controlling and measuring voltages across spans of rails, rail-to-earth voltages, and the DC current source output. The stray current collector cable shall be monitored for electrical continuity of the stray current collector cable. Electrical continuity shall be monitored by controlling and measuring a DC current source output and the associated collector cable currents.

Term	Definition
Stringing	Installation of overhead wires under tension.
Structure	A principal support for the OCS conductors, normally including foundation, pole(s), and cantilever(s)/bracket arm(s) or head-span/cross-span; or eyebolts and cross-span.
Support Bracket for Bar Insulator	Part of the Anchor Bracket Assembly. This assembly attaches the bar insulator into the track support weldment. The threaded rod is a provision for height adjustment.
Support Clamp Weldment	Part of the Anchor Bracket Assembly. This weldment holds the vertical support in place against the corner of the concrete beam.
Surge Arrester	A device that protects other electrical equipment by “arresting” or discharging surge currents brought about by external (e.g. lightning) or internal (switching events) forces.
SCAT	Simple Catenary Auto-Tensioned (SCAT) overhead contact system.
SWAT	Single Wire Auto-Tensioned (SWAT) overhead contact system.
SCFT	Simple Catenary Fixed Termination (SCFT) overhead contact system.
SWFT	Single wire fixed termination (SWFT) overhead contact system.
Tension Section	Length of OCS between two corresponding terminations with automatic tensioning or fixed terminations.
Tensioning	A method of controlling sag in an OCS by tension or by weight.
Tensorex C+ Assembly	Tensorex C+ are spiral spring tensioning devices used in auto- tension catenary system. Tensorex C+ exert a constant tension on the OCS Contact or Messenger wires regardless of any changes in wire length due to temperature. The special spiral springs are firmly connected to two cams on the same shaft. The degree of rotation of the shaft results in a linear variable torque/moment, which is compensated for by the combination with the variable radius of the cam. This produces a constant pulling force over the entire device working range.
Termination Assembly	Part of the Expansion Joint Assembly. Holds the OCR transition bar.
TOR	Top of rail (as a vertical datum).
Track Support Weldment	Part of the Anchor Bracket Assembly. This weldment carries the bar insulator and the OCR bars on both sides.

Term	Definition
Track Tolerance	• Vertical - allowable variation in track height. • Lateral - allowable variation in the track alignment. • Cross level - allowable variation between the levels of the two rails (or the designated difference in levels on super-elevated track).
Traction Electrification System (TES)	The entire traction electrical distribution system from the power supply substations to the light rail vehicles, including power supply ducts, feeder cables and risers, and the OCS.
Traction Power Chief (Chief)	Person who supervises Rail Electrical Worker Crew Chiefs, Leads, Rail Electrical Workers, Apprentices, and Helpers to operate, maintain and repair the Link Light Rail Traction Power Electrification System.
Traction Power Substation (TPSS)	A facility also known as a substation used to convert AC voltage to DC voltage for distribution to the OCS.
Trolley Wire	A solid wire, usually grooved, carrying power from which trolley poles of buses pick up electrical power (see also “contact wire” for LRVs).
Tunnel Support	A catenary, trolley wire, feeder, or return wire support for use in tunnels normally mounted in the tunnel soffit.
Uplift	The difference in height of contact wire when at rest and when subjected to an upward force under one of the following conditions: • Dynamic - due to the pantographs passing. • Static - due to stationary pantographs.
Vertical Support Weldment	Part of the Anchor Bracket Assembly. This weldment comprises the main support at the front and rear side of the anchor bracket.
Wire Run	An automatic tensioned or fixed termination wire tension segment of an overhead contact system.
Yoke Plate	A steel plate or casting on which two or more wires terminate on one side and continue as one wire on the opposite side (usually yoke-shaped and with lever action to distribute loads from a counterweight to the messenger and contact wire).

3.0 PROCEDURAL REQUIREMENTS

3.1 Roles and Responsibilities

i. **King County Metro (KCM)** is accountable for maintenance, repair, and inspection of Sound Transit equipment. King County Metro is responsible to:

a. Maintain list of qualified persons and provide to Sound Transit for approval.

b. Perform OCS maintenance, repair, and inspection in compliance with this Maintenance Standard.

c. Inspect work performed by King County Metro MOW, contractors, and non-agency parties before placing equipment into service.

d. Ensure qualified personnel conduct the maintenance procedures.

e. Maintain inspection records and forms within EAMS.

f. Report to Sound Transit existing OCS asset conditions in accordance with standards and requirements and document conditions within EAMS.

g. Take appropriate action as specified in this Maintenance Standard.

h. Request approval from Sound Transit Operations and collaborate in review of requests before deviating from OCS requirements or this Maintenance Standard.

i. Maintain third party inspection reports provided by Sound Transit in EAMS.

j. Provide inspection and maintenance records to Sound Transit.

k. Provide appropriate training for its qualified personnel.

ii. **Sound Transit** is responsible to:

a. Periodically update this Maintenance Standard.

b. Audit inspection and maintenance records to ensure conformance with this Maintenance Standard.

3.2 Qualified Persons

KCM must appoint and maintain a list of qualified persons to oversee OCS maintenance, and inspections, and to inspect the work of non-agency parties. KCM and non-agency parties shall abide by Sound Transit's Electrical Safety Program and ensure that:

i. A list of qualified persons is maintained and provided to Sound Transit for approval.

ii. A system is established for training, evaluating the credentials of, and appointing qualified persons to ensure they have sufficient knowledge of the equipment and its hazards before performing work on such equipment.

iii. Employees receive an appropriate level of training before being designated a qualified person.

The following list defines the types of qualifications required to be obtained for specific types of work:

iv. **Employee in Charge (EIC):** EICs shall be designated KCM employees who hold a current KCM Light Rail EIC certification and oversee a work group. EICs may be assisted by flaggers when necessary.

v. **Rail Electrical Worker (REW):** REWs are qualified electrical employee assigned to the Traction Power group. REWs shall be knowledgeable QEEs who have demonstrated additional expertise working with high voltage electrical systems.

vi. **Qualified Electrical Employee (QEE**): QEEs shall be familiar and knowledgeable in the construction and operation of the electric power generation, transmission, and distribution equipment related to OCS, and such lines and/or equipment that concerns their position. OEEs shall be fully aware of potential hazards related to performance of the work, and who has passed a journey status examination for the branch of the electrical trade with which they are connected.

vii. **Traction Power Chief (Chief):** Person who supervises Rail Electrical Worker Crew Chiefs, Leads, Rail Electrical Workers, Apprentices, and Helpers to operate, maintain and repair the Link Light Rail Traction Power Electrification System

3.2.1 Work Performed by Others

Work performed by contractors or other non-agency parties must be inspected by a qualified person prior to placing the equipment into service. All requirements of KCM in relation to compliance with standards and maintenance of records are applicable to Contractors. Source control of material supplied by Contractors and up-to-date calibration records of tools used by Contractors are required.

3.3 Operational Records

Sound Transit requires operational records for maintenance tasks. All changes to Sound Transit's systems/equipment are to be reviewed and approved by Sound Transit Configuration Review Board (CRB).

3.3.1 Single Line Diagrams (SLDs)

Sound Transit's Electrical Safety Program requires SLDs to be maintained, including those for sectionalizing.

3.3.2 Configuration Files

Each software configuration file must be maintained for all configurable devices and tracked for any field modifications.

3.4 Incident Logging

To track events in the field involving Sound Transit assets and equipment, written logs are to be maintained in the main electrical room or in the service entrance panel for each facility. The following process to record and track events is required:

i. ST Facilities Manager gathers incident information or coordinates with others (e.g. technicians, SOC, etc.) to gather and input the following information into EAMS:

   a. Date/Time/Duration of incident.
   b. Name of Persons and their organizations supporting troubleshooting
   c. Description of incident, including but not limited to:

- Potential reasons for the event.
- Equipment impacted.
- Panel & Circuit Breaker #, if applicable.

ii. Complete an Engineering Service Request (ESR) if engineering support is needed for troubleshooting. Input EAMs number into ESR description.
iii. For incidents involving downtime or significant cost impacts, ST AO Engineering to conduct an incident investigation including a Root Cause Analysis and generation of a report.

3.5 Single-Point Data Consolidation and Trending Required

Single-point data consolidation and trending requirements will be updated in future revisions of this Maintenance Standard. Recommendations for data consolidation are included under Section 8.0 OCS Preventive Maintenance Tasks Table.

3.6 **Priority Rating** (Based on APTA-RT-FS-S-006-03, Rev. 2 Section 8, SMP's, and EAMS)

The rail transit system shall develop a priority rating system to evaluate and determine the effects that any single defect will have on the system if it chooses to operate with a known defect.

The following are the recommended defect priority ratings:

- **Priority 1 Defect (EAMS Priority ID A):** Defects whose failure would cause immediate service disruption, significant safety hazard.
  Action: A permanent or temporary repair shall be made immediately.
  Return to Service – Upon completion of the priority 1 defect repairs refer to SMP 28.8 Power Interruptions and Return to Service, prior to returning to revenue service.

- **Priority 2 Defect (EAMS Priority ID B)** - The defect may cause disruption of revenue service or defects which degrade reliability or increase risk but are not imminent.
  Action: schedule repair within next or off-peak window.

- **Priority 3 Defect (EAMS Priority ID C)** - The defect will not affect revenue service. Such designation alerts to an OCS or TPSS condition that may degrade to a worse condition if left uncorrected or defects which are non-critical today, have low impact, but should be monitored or included in the next capital/maintenance batch.
  Action: Correct if possible or monitor, plan for repair in next cycle. Uncorrected defects shall be recorded, and the reports shall be used for scheduling future work.

4.0 SYSTEM OVERVIEW

Sound Transit's Link Light Rail uses an Overhead Contact System (OCS) to supply electrical power to LRVs operating throughout the system. The OCS consists of overhead conductors that are energized at 1500 volts DC fed by Traction Power Substations (TPSS) located approximately every 1 to 1.5 miles on the alignment. LRVs pull power from the overhead conductors using collectors called pantographs that are affixed to the top of each vehicle. The circuit is completed by a return path through the running rail and back to the TPSSs via negative return cables.

The OCS consists of all equipment electrically connected between the feeder cable terminations and the vehicle pantograph, as well as all the mechanical systems required to physically support the overhead conductors.

Using DC feeder breakers, switches, and section insulators, the OCS is sectionalized into power sections, which can be de-energized manually, remotely via SCADA, or automatically through protective devices at the substations. Each power section is monitored by protective relays on the DC feeder breakers as well as TPSS AC main breakers, which will automatically trip the breakers and de-energize the power sections under certain safety critical situations, such as an OCS short circuit.

4.1 OCS Technical Specifications

Sound Transit's OCS is designed around the following technical specifications:

i. Overhead Conductors:

   a. 350 KCMIL hard-drawn solid grooved copper contact wire, 3300-lbs normal tension at nominal ambient temperature of 60°F.
   b. 500 KCMIL hard-drawn 19-strand copper messenger wire, approximately 5000-lbs normal tension at nominal ambient temperature of 60°F and dependant on span length per conductor temperature/ tension charts.

ii. Voltage Rating:

   a. Two levels of insulation minimum between energized parts and grounded structures.
   b. Each level rated to be compatible with system insulation requirements.

iii. Overhead Contact System Types and Locations:

   a. Simple Catenary Auto-Tensioned (SCAT).
   b. Single Wire Auto-Tensioned (SWAT).
   c. Simple Catenary Spring Tensioned (SCST). Spring Tensioners included on the I-90 Floating OCS/OCR.
   d. Single Wire Fixed Termination (SWFT).
   e. Low Profile, Simple Catenary Auto Tensioned (LPSCAT).
   f. Low Profile, Simple Catenary Fixed Terminated (LPSCFT).
   g. Overhead Conductor Rail (OCR).

iv. Open-route Mainline Alignments without Low -Clearance Overhead Bridges:

   a. Simple Catenary Auto-Tensioned (SCAT).

v. Tunnelled Mainline Alignments:

   a. Low Profile, Simple Catenary Fixed Terminated (LPSCFT).

vi. Open-Route Mainline Alignments with Low-Clearance Overhead Bridges:

   a. Low Profile, Simple Catenary Auto Tensioned (LPSCAT).
   b. Overhead Conductor Rail (OCR).

vii. Street-Running Low Speed Alignments:

   a. Storage Yards
   b. Operations and Maintenance Facilities:

viii. Overhead Conductor Rail (OCR).
ix. Climactic Conditions Table 4-1 (ST Requirements Manual):

Table System Overview- 2 Climatic Conditions

Condition Type	Conductor Temperature	Radial Ice Accumulation	Wind Speed
Operating Condition 1	60 degrees F	None	None
Operating Condition 2	0 degrees F	None	55 mph
Operating Condition 3	0 degrees F	Contact Wire: 0.25 inches Messenger Wire: 0.5 inches	40 mph
Non-Operating Condition 1	0 degrees F	None	70 mph
Non-Operating Condition 2	32 degrees F	Contact Wire: 0.5 inches. Messenger Wire: 0.5 inches	None
Temperature Range of Conductors for Auto-Tensioning	5-130 degrees F	N/A	N/A

4.2 Feeding Arrangement

The system is designed with an “N minus 1” contingency scenario, so that if a minus one condition develops where a single TPSS becomes out-of-service, the adjacent TPSS' can typically provide full operational capability. This allows the system to maintain revenue operations at full capacity with OCS disconnect switches configured to bypass one substation.

In the situation where a TPSS is offline and not configured for bypass operations, adjacent TPSSs will be single end fed and may require reduced operations.

4.3 Unique OCS Features

4.3.1 I-90 Floating Bridge

On the I-90 Floating Bridge over Lake Washington, steel portals, spaced at a maximum of 177-feet. I-90 also has Overhead Conductor Rail (OCR) expansion joints at each end of the bridge known as the CR-EXP assemblies, in addition to Simple Catenary Auto-Tensioned (SCAT) OCS. Cantilevers, cross spans, and balance weights attached to the steel portals hold the messenger and contact wire in place and maintain a contact wire height per OCS Layout Drawings and Acceptance Measurement from the Top of Rail (TOR).

On the I-90 Floating Bridge, the negative return rail is designed with a remarkably high path of resistance, 1,250 ohms - 1000 track feet, between the rail and the bridge deck. This design helps minimize stray current and ensure that the rail acts as the lowest path of resistance back to the substations. Also, an improved bridge cathodic protection system and stray current collector mat with monitoring system is integrated with the track design to mitigate any remaining stray current.

4.3.2 Electromagnetic Interference (EMI) Section from University of Washington Station (UWS) to U-District Station (UDS)

As required by the University of Washington campus in Seattle, the traction electrification design is required to employ mitigation measures to reduce the levels of stray magnetic fields within the vicinity of sensitive university equipment. A “Hi-Lo B-field mitigation” strategy for electrification distribution along with rail-to-ground monitoring is employed to satisfy the requirements. The mitigation uses an insulated positive parallel feeder between the tracks under the floating slabs to carry the current and is frequently connected to the contact wire via riser taps which are approximately every 66 feet along-track. A non-current carrying messenger wire is used for vertical support of the contact wire only.

5.0 COMPONENTS AND ASSEMBLIES

5.1 Insulated Overlaps

Insulated overlaps are the primary means for electrical sectioning and isolation; creating discrete sections to enable TPSS protective relays to disconnect faulted sections and minimize disturbance to the remaining system, achieve flexible operation during system emergencies, meet AHJ requirements for fire zones in tunnels, and to enable safe performance of maintenance. Sectioning of the OCS may also be provided through section insulators.

5.2 Uninsulated Overlaps

An overhead contact system (OCS) overlaps without sectionalizing. A construction overlap is required to avoid possible excessive stagger change, due to excessively long tension lengths of typically auto-tensioned (AT) wire.

5.3 Tensioning Devices

Conductors must be tensioned to have the appropriate sag between supports. Tensioning devices are used to apply the required line tension into the conductor and safely transfer the tension to a supporting structure. Tensioning devices include conductor termination clamps, line insulators for double insulation, and turnbuckles for maintenance adjustments. Tensioning devices may employ means to accommodate variation of along-track movement or they may have fixed termination lengths. The mainline system is mainly auto-tensioned (AT) in open-route areas. Other areas such as tunnels including University Link, DSTT, Beacon Hill, Downtown Bellevue, and other locations are SCFT (Fixed Termination) tension lengths.

5.3.1 Balance Weight Assemblies (BWA)

The primary method of maintaining constant conductor tensions in existing alignments, including 1 Line Central Link, Angle Lake, Northgate, and 2 Line East Link Extensions. Concentric style tension wheels with a 3:1 ratio between line tension and weight stack. Weight stacks can be either a single billet of steel or made up from multiple steel plates each ranging between 50 and 65 lbs. Solid weight stacks may be positioned within a round tubular pole in areas accessible to the public. Weight stack travel limiting brackets are used for both high and low temperature stops.

5.3.2 Constant Tension Spiral Spring Tensorex C+

The primary method of maintaining constant conductor tensions on the latest expansions including 1 Line Lynnwood Link, Federal Way, and 2 Line Downtown Redmond Link Extensions as well as future alignments. Spiral wound spring with cam connection for compensating standard tension lengths. Two independent units per conductor.

5.3.3 Variable Tension Linear Spring

Linear compression or box spring for short tension lengths in yard or for crossover wire runs. These are also used on the I-90 floating bridge expansion joints.

5.3.4 Fixed Termination

Direct fixation of conductor to termination structure. Used either in storage yard and operations and maintenance facilities or paired with an auto-tensioning device for a half-tension length. These are also used in mainline tunnel runs (University Link, Northgate Link, Central Link DSTT, Beacon Hill, Downtown Bellevue, and others).

5.4 Section Insulators

Section insulators are used for electrical sectioning and isolation where space is limited and an insulated overlap may be unpractical, such as in crossover tracks or in the storage yards.

5.5 I-90 Floating Bridge OCR Expansion Joints

The I-90 floating bridge structural system contains a flexible transition segment between land and water. The trackwork as well as OCS are designed to accommodate extreme movement of the joints of the transition segment. The CR-EXP assembly is a custom designed and fully integrated system of components to safely handle the required movement while supporting uninterrupted service. Due to the complexity of the equipment, the manufacturer’s specific maintenance instructions and drawings should be referenced directly.

5.6 I-90 Floating Bridge Catenary Monitoring System

Balance weights on the I-90 floating bridge are fitted with a monitoring system that evaluates changes to the conductor tension and alerts through SCADA if thresholds have been exceeded. The system relies on non-contact sensors attached to the balance weight wheels, data acquisition stations that gather the data from multiple sensors, and an evaluation station that trends the data and sends the alerts to SCADA. Due to the

complexity of the equipment, the manufacturer's specific maintenance instructions and drawings should be referenced directly.

5.7 Midpoint anchors

Fixing of along track movement at or near the midpoint of each full tension length is accomplished by midpoint anchors. Anchor provides means of restricting movement for both the contact wire as well as messenger wire.

5.8 OCR

Contact wire supported from and securely held within an extruded aluminum conductor rail profile may be used in low clearance areas. Conductor rail reduces the required headroom necessary in tunnels. The contact wire held without line tension for much of the system outside the first 3 to 5 elements. Changing from flexible system to OCR is accomplished by a special transition element. The conductor rail is expected to have along-track movement, and the system is therefore outfitted with overlaps and midpoints like a flexible system.

5.9 Crossovers

Wiring above special trackwork is generally of the crossing design. Contact wire bridges are used to prevent pantograph fouling and electrical jumpers are used to maintain equal potential.

5.10 Supports

The vertical weight, lateral radial loading, and/or line tension of the OCS conductors periodically must be transferred to ground. This is accomplished by several types of supports depending on which of the loads are being supported and how they can be transferred to ground.

5.10.1 Poles

Poles are the primary form of transferring the OCS loads occurring at a vertical height above rail down to grade. Free standing poles are typically tapered tubular or wide flange. Poles are typically fitted with baseplates and coupled to a concrete pier by anchor rods and double nut without grout.

5.10.2 Drop Posts

Where supports are required, but poles with footings are not possible, drop posts attached to overhead or lateral structures can be used. They generally have the same

requirements as OCS poles; however, the overall load requirement is less due to the reduced vertical height from conductors to attachment.

5.10.3 Portals

Where supports are required, but ground conditions require lateral loads to be reduced, portal structures can be used. They can be fitted to concrete piers in the same manner as poles or attached to existing structures.

5.10.4 Cantilevers

Cantilevers are the primary form of transferring the OCS loads occurring above track to a horizontal offset outside the vehicle envelope. Cantilevers are made from round steel tubing and provide direct rigid attachment of the conductors. Cantilevers are supported from poles, drop posts, or portals from hinged brackets to accommodate along-track movement.

5.10.5 Cross Spans

Cross spans are used to transfer the loads horizontally away from the track centerline where the required offset is generally too far for cantilevers or where multiple tracks may be required to be supported from common structures. The OCS is supported from flexible steel cable strung between two structures of either pole, drop post, a portal structure.

5.10.6 Hangers

Contact wire is vertically supported at frequent intervals from messenger wire by hangers such that the contact wire is nearly straight in reference to the rail. Hangers are non-current carrying and are made from flexible stainless wire.

5.10.7 Grounding

All supporting structures shall be independently and solidly grounded for personnel protection and bonded to prevent stray current corrosion.

5.11 Surge arresters

Surge arresters located at disconnect switches protect the underground insulated feeder cable. Surge arresters also located at tunnel portals or low clearance structures protect against overvoltage arcing.

5.12 Maintenance and Storage Facilities

i. Door bridges

Door bridges allow operation of pantographs through vertical rolling doors at the face of maintenance facilities. While the rolling doors are open, the door bridge provides a continuous path for the LRV to travel in/out of the facility under pantograph power. While the rolling doors are closed, the door bridge provides for the separation and required electrical clearances by way of a vertical swing-away arm.

ii. Yard to mainline lead tracks.

Mainline tracks typically are simple catenary auto-tensioned while yard tracks are typically single wire fixed termination. The transition between these two systems may require additional special inspections.

6.0 CONDITION ASSESSMENT

Condition assessment is a key component in a Transit Asset Management Plan. The overall condition rating will become the State of Good Repair (SGR) score for an asset. Knowing the condition of a Traction Power assets is essential both to ensure safe and reliable service daily and to optimally prioritize future capital investments. This section describes the methods used to assess conditions for the purposes of long-range investment planning and provides summary data of current conditions by asset class.

6.1 Condition Rating Methodology

The following Transit Asset Condition Assessment Methodology is used to assess the condition and monitor the performance of assets using the Federal Transit Administration (FTA)'s scale:

iii. Onsite visual, physical condition assessments, using the FTA 1-5 Scale:

a. The asset's physical condition method refers to developing a score based on the observed condition of the asset, by visual assessment of the asset that may include exterior and interior condition, defects, etc. Existing periodic inspections may perform such inspections; where feasible, condition scores may be derived from such inspection reports or depending on the asset types.

iv. Maintenance condition using inspection data and/or age, on the FTA 1-5 scale:

a. The condition uses the total preventative maintenance activities performed to arrive at a condition score when applicable. Previous maintenance inspection data can also be leveraged to derive a more accurate maintenance condition score.

v. Reliability Condition using Deficiency Record and history, on 1-5 scale:

   a. This condition uses the Asset's history of deficiencies and work order repairs over time to determine its failure rate. A high failure rate indicates that an asset is less reliable and more likely to fail, which can lead to higher maintenance costs and increased downtime. An asset's failure rate changes over its lifecycle. When an asset is new, its failure rate is at its lowest. Criteria are reserved and intended for future use once all OCS assets and inventory data are implemented within EAMS.

vi. Useful Life Benchmark (ULB):

   a. The ULB is defined as the expected lifecycle of a capital asset for a particular asset in service, or the acceptable period of use in service for a particular Traction Power substation. In most cases, if an asset exceeds its ULB, then it is a strong indicator that it may not be in a state of good repair. Asset age is one of the primary variables typically used for determining remaining useful life because, based on the age of an overall asset, metrics are used to determine the percentage of the age that has been consumed.

6.2 Condition Assessment Measure TERM Scale

The condition assessment scoring methodology detailed in this section uses a five-point scale developed by the FTA's Transit Economic Requirements Model (TERM) scale. An asset is deemed to be in good repair if it has a rating of 3, 4 or 5 on this scale. Likewise, an asset is deemed to not be in good repair if it has a rating of 1 or 2. A summary of the condition assessment criteria for each methodology is shown in Table Condition Assessment-3. An example of this assessment being used on an OCS asset is shown in Table Condition Assessment-4. To subsequently determine the overall health of an asset, an alternative index weighting scale, shown below, will be used. If a critical part of an asset is deemed obsolete and/or it is not available anymore, then its overall condition assessment score will be 1 regardless of the other condition criteria.

Table Condition Assessment-3 Condition Assessment Matrix

Condition Assessment Criteria				Rating Scale	
Visual Condition	**Maintenance Condition**	**Useful Life Benchmark (ULB)**	***Reliability Condition****	**Rating**	**Rating Description**
Annual scoring during maintenance	Calculated through PowerBi and EAMS		*Reserved for future implementation*		
No visible defects, near new condition, may still be under warranty if applicable.	Meets or exceeds <1% of preventive maintenance activities deferred.	Asset is new or nearly new with 75%-100% ULB age remaining	*Asset is recently put in service and/or shows no signs of wear.*	5	Excellent
Good condition, but no longer new, may have some slightly defective or deteriorated components(s), but is overall functional.	Nearly meets; 1% to 10% of preventive maintenance activities deferred.	Asset is nearing or at its mid-point of ULB with 50%-75% age remaining	*Asset is performing as its intended function. Any deficiencies are normal wear. Assets need some minor repairs for subcomponents.*	4	Good
Moderately deteriorated or defective component(s); but has not exceed its useful life.	Does not meet 11% to 20% of preventive maintenance activities deferred.	Asset has passed its mid-point of ULB with 25%-50% age remaining	*Low Failure rate. In the early stages of affecting performance, operation, and maintenance. Assets require more frequent minor repairs on subcomponents*	3	Adequate
Defective or deteriorated component(s) in need of replacement, exceeded useful life.	Does not meet; 21% to 30% of preventive maintenance activities deferred.	Asset is nearing or at its end of ULB with 0%-25% age remaining	*Medium Failure Rate. Performance, operation, and maintenance is significantly affected. The effects are large and continuing to increase. Significant increase in repairs between preventative maintenance cycles*	2	Marginal
Critically damaged component(s) or in need of immediate repair; well past useful life.	Does not meet targeted preventive maintenance activities. Is non-operable.	Asset has gone beyond its ULB age	*High Failure rates. Performance, operation and maintenance is severely affected. Equipment is approaching failure or is completely failed.*	1	Poor

*Criteria currently not implemented in condition assessment score

Table Condition Assessment-4 Condition Rating Matrix Example

ASSET CLASS	ASSET TYPE	VISUAL CONDITION	MAINTENANCE CONDITION	USEFUL LIFE BENCHMARK	CONDITION RATING
OCS	OCS Tapered	2	5	1	3

ASSET CLASS	ASSET TYPE	VISUAL CONDITION	MAINTENANCE CONDITION	USEFUL LIFE BENCHMARK	CONDITION RATING
	Tubular Pole				
	OCS Wide Flange Pole	2	5	1	3
	OCS Down Guy Anchor	2	2	3	2.35
I-90 Floating Bridge Systems	OCS Portal Assembly	1	1	5	2.4
	Portal Hinge Base	3	2	5	3.25
	Expansion Joint Assembly (CR-EXP)	2	1	5	2.6
	Anchor Bracket Assembly	2	1	5	2.6
	SCCCRMS	5	4	3	3.85
	Jumper Assembly	4	4	2	3.3
OCR	Conductor Rail Bridge Expansion Assembly	3	4	4	3.8
	Tunnel OCR	3	4	4	3.8

Table Condition Assessment- 5 Condition Rating Index Weighting

VISUAL CONDITION	MAINTENANCE CONDITION	USEFUL LIFE BENCHMARK	RELIABILITY CONDITION	OVERALL CONDITION RATING
20%	45%	35%	T.B.D.	100%

6.3 Condition Assessment Reporting

The visual condition assessment will be scored and inspected to determine an aspect of the asset condition on an annual basis. While completing the scoring, the inspector should be looking at the overall visual condition of the asset, including any loose, damaged, leaking, or deteriorated items of the asset. The other elements of Maintenance, Reliability and ULB condition scoring will be calculated in EAMS, informed from maintenance records, deficiency or repair records and age of asset remaining.

7.0 FUNCTIONAL TESTING

This Maintenance Standard establishes minimum standards for functional testing to ensure the OCS end to end functionality such as the pantograph OCS interface. Requirements for functional testing will be elaborated further in future revisions to this document. Reference sample OCS Test Procedures attached with this standard in Appendices 10.4 and 10.5.

7.1 Cables

7.1.1 Visual and mechanical inspections (per APTA-RT-FS-S-006-03 Rev 2 Section 6.2)

The rail transit system shall perform the following inspections on all positive and negative cables when cable failure occurs in areas such as conduits, duct banks or troughs.

a) Inspect exposed sections of cables for physical damage or evidence of overheating.

b) Inspect fireproofing in common cable area and firewall penetrations, if applicable.

c) Inspect terminations, jumpers, and splices for evidence of physical damage or overheating.

d) Inspect bolted electrical connections for high resistance using one of the following methods:

- Use the calibrated torque-wrench method in accordance with OEM published data to verify tightness of accessible bolted electrical connections.

- Perform a thermos-graphic survey of the equipment under loaded conditions.

e) Inspect the cable support and termination.

7.1.2 Electrical Tests

a) Perform resistance measurements on bolted connections with a low-resistance ohmmeter capable of reading 2-μΩ, (micro-ohms) or with parallel cables use a clamp-on ammeter to verify nearly equal currents on all cables.

b) Perform an insulation resistance test, only when the integrity of a cable is suspect, with a megohmmeter using a voltage no greater than the cable insulation rating. Where there is more than one cable in parallel, test each individual cable.

7.1.3 Test Values

a) Compare bolted connection resistances to values of similar connections.

b) Ensure that bolt-torque levels are in accordance with OEM recommendations.

c) Ensure that micro-ohm or millivolt drop values do not exceed the high levels of the normal range as indicated in the original equipment manufacturer (OEM) published data. If OEM data is not available, then investigate any values that deviate from similar connections by more than 25 percent of the average value.

d) Compare test results to previously obtained results.

7.2 For additional testing, reference sample OCS Test Procedures included and attached with this OCS Maintenance Standard in Appendices 10.4 and 10.5.

8.0 PREVENTATIVE MAINTENANCE TASK TABLE

Listed below are the preventive maintenance tasks of OCS Equipment. These tasks have been informed by the official O&M recommendations per the product manufacturer/installer and existing maintenance tasks. The tables below establish a recommended standard for periodic visual, electrical, and mechanical inspections, as well as necessary tests to be performed.

8.1 OCS Maintenance Task Table

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
Foundations	Condition of concrete, visible cracks, deposits of rubbish, overgrowth of vegetation, concrete condition, and pooling water.	Annually	Y – Monitor and record locations with worsening conditions. Increase inspection cycle, as necessary	OCS-001
Poles	Check poles for corrosion, loose nuts on foundation bolts, rake and bending of poles – particularly termination poles.	Annually	Y – Monitor and record locations with worsening conditions. Increase inspection cycle, as necessary	OCS-002

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
Connections – Feeder Cable, negative return cable, Jumper cable, and connections to surge arresters	• Inspect bolted electrical connections, when deemed necessary, for high resistance using one of the following methods, and record which of the following methods is used: • Insulation resistance test: Perform with a megohmmeter using a voltage no greater than the cable insulating rating. Since this test requires unbolting the cables, it should be performed prior to checking bolt torque, or connection resistance check, and may as an alternate perform thermographic survey. • Thermographic survey: Perform on the equipment under loaded conditions. Record location and value of highest temperature. • Resistance measurements: Perform on bolted connections with a low resistance ohmmeter capable of reading 2 μΩ, or with parallel clamp-on ammeter to verify nearly equal currents on all cables.	Annually	N	OCS-003

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
	• Insulation resistance test: Perform with a megohmmeter using a voltage no greater than the cable insulating rating. • Compare bolted connection resistance to recorded values or values of similar connections. Investigate any values that deviate from similar connections by more than 25% of average value. • Use a calibrated torque wrench to verify tightness of bolted electrical connections. Initial existing torque marks may serve as a starting point but are not always reliable, random checks should be performed and when a loose connection is found, all bolted connections at location should be checked for compliance. • Ensure bolt-torqued levels are in accordance with OEM recommendations. • For an OCS circuit issue, perform OCS Circuit and Continuity Loop Resistance Test, to troubleshoot and narrow down the problematic area for remediation.			

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
I-90 Floating Bridge Portal Structures Base Plates	• Check resistance of base plates, recommend a random sample set and results recorded, alternating locations as these are performed in the future. • Measure between anchor bolts and baseplate to get resistance value. • Resistance of 25-ohms would be ideal.	Annually	N	OCS-004
I-90 Floating Bridge Aerial Ground	• Check resistance of grounding cable. • Resistance of grounding cable to be approximately 0.30-ohms for length of bridge or 0.150-ohms from center of the bridge to each TPSS ground mat. • Record measurements and investigate and repair any measurement 50% over nominal. • Check cables for strand breaks, continuity, and integrity.	Annually	Y – Monitor and record locations with worsening conditions. Increase inspection cycle, as necessary	OCS-005
OCS-to-Ground	• Check resistance of grounding cable. Resistance to be 25-Ohms maximum for OCS poles and supports.	Annually	Y – Monitor and record changes in resistance.	OCS-006

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
Traction Power System Grounding & Negative Return Cables Connections	• Check positive feeder cables and negative return cables for strand breaks, continuity, integrity of all grounding cables, and possible damage of bonding. • Check the negative return cable connections to impedance bonds.	Annually	N	OCS-007
Warning Signs, Pole Number Nameplates/ Signage	• Check for the proper presence including any damaged, lost, or loose signage. • Check the cleanliness and firm attachment of signage within Right-Of-Way, including all warning signs, Milepost signs, and OCS Poles numbering nameplates.	Annually	N	OCS-008
Protection Guards / Covers	• Check protection guards and covers for corrosion damage, firm attachment, and correct positioning.	Annually	N	OCS-009
Insulators and Cable terminations	• Check for damaged fiberglass or epoxy and contaminated insulator surfaces. • Clean OCS Insulators.	Annually	N	OCS-010

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
Contact Wire (CW) Wear	• Measure and record contact wire wear taking reading from top to bottom of the wire, typically using digital caliper tool, periodically calibrated, as necessary. Initial diameter of 350 MCM Contact Wire used in ST system is 0.620-inches diameter. • Document location of measurement. Document locations of contact wire-wear greater than 18% or 0.483-inches. • Take note of any locations showing signs of arcing and/or excessive wear. Verify wire wear in those locations. • Once wire measurements are regular, wear rates can be established for each location and planning for replacement should be done based on this data. o CW found between 20% (0.475”) and 25% (0.450”) worn shall be flagged for more regular inspections.	Annually	Y – Monitor and record measurements at locations with concern of wire wear. Increase inspection cycle accordingly when approaching excessive wear.	OCS-011

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
	o CW found worn more than 28% (0.434") submitted for replacement, such that it is replaced prior to reaching 30% (0.424"), which is the maximum threshold. o CW wear of 30% (0.424") is the limit and should be flagged for immediate follow-on action (splicing out) if full wire run or segment replacement is planned for later in the future.			
Contact Wire (CW) Acceptance Measurements Verification	• Measure/ Record Contact Wire Height & Stagger. • Measurements taken from centerline of track, Top-of-Rail (TOR), at OCS supports and mid span.	Annually	N	OCS-012
Cantilever	• Check the along track position of the cantilevers (according to current temperature and along track movement charts, found in the Contract Drawings) and the inclination of steady arms considering temperature and distance from mid-point. Best to perform	Annually	N	OCS-013

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
	inspection when weather near nominal design temperature (60 degrees F).			
Catenary	• Check the catenaries for broken/kinked single wires, position of contact wire bridges at crossovers depending on the temperature and length to fixed or midpoint anchor, slackened, non-plumb or missing jumpers, position of wires at overlaps, crossovers, and evidence of arcing at section insulators.	Annually	N	OCS-014
Hangers	• Check the hangers for loose, torn, or displaced strands, proper alignment and fit of the clamps and damaged contact surfaces.	Annually	Y – Monitor and record locations for replacement with poor conditions if not replaced during Preventive Maintenance.	OCS-015
Tension Wheel Assembly	• The free movement of the tension wheel must be checked by lifting the weight column. The weights must	Annually	N	OCS-016

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
	return into their original position when left alone again within ± 4 inches. • The wire wrap and weight stack position should be checked against nomograph, or alignment chart, provided in the contract assembly drawing. • Check wire tension for constant tension springs against manufacture's data. • All components in arrangement must also be checked for corrosion and breakage damage. Faulty or damaged components need to be replaced. • Check that appropriate fittings are greased according to manufacturer's specifications.			
Yoke Plate	• Check the vertical position of the yoke plate which should be near plumb. • Check the yoke plate for proper alignment and no loose components.	Annually	N	OCS-017

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
Pole Mounted Disconnect Switches	• Check the pole mounted switches for the condition of the contacts, condition and position of the knife-blade contacts, arcing horns, and connection cables. • Check condition of the linkage, pipes, and handle, and outside condition of the switch mechanisms. Operate the switch using the operating handle to check for smooth operation. • Check for proper operation of MOD's (Motor Operator Disconnects), where applicable. • Verify and record correct Position Limit Switches indications back to TPSS LCMS and via SCADA to Link Control Center (LCC).	Annually	Y – Monitor and record locations with OCS pole-mounted disconnects having incorrect indications and any worsening conditions. Increase inspection cycle, as necessary	OCS-018
Section Insulators (SI's)	• Check section insulators for arc free or minimal arcing operation during the passing of operation trains. Reference Live Wire testing documents and procedure. • Check Height, Stagger, and wear. Stagger for SI's ideally centerline 0" Stagger, max Stagger +/- 2 inches from	Annually	Y – Monitor and record locations with worsening conditions. Increase inspection cycle, as necessary	OCS-019

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
	centerline. For any locations higher than +/- 2 inches recommend increase inspection frequency. • Clean Section Insulators. • In accordance with manufacturer's specifications, check for smooth transition across section insulator using the appropriate level acting as the pantograph.			
Electrical Connectors	• Check the electrical connectors for broken single strands, damaged contact surfaces, and tight fit of the clamps. • Check Torque per manufacturers recommendations.	Annually	Y – Monitor and record locations for repair with poor conditions if not performed during Preventive Maintenance.	OCS-020
Wire Run Inspection	• Visually inspect and torque all attachments per manufacturers recommendations. • Clean and inspect each insulator. • Inspect all feeder taps and clamps. • Inspect bonding of potential jumpers.	Annually	Y – Monitor and record locations for repair with poor conditions if not performed during Preventive Maintenance.	OCS-021

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
	• Inspect all hangers and clips. • Inspect contact wire for kinks, bends, twists. • Inspect for wear and arcing locations, and evidence of hard spots, a fish scale can be useful for uniform hanger (dropper) tension.			
External Lines (Telecommunications, Signal and Other Lines)	• Check the clearance between external lines and the overhead contact system lines. Required clearances will vary according to type of external lines and voltages. • Clearance typically no less than 4 feet for lower voltage lines. • Utility high voltage distribution, transmission lines will require much greater clearances, according to voltage.	Annually	N	OCS-022
Return Cable Connections	• Check the return cable connections for mechanical damage and tight fit of the cable lugs. • Perform visual inspection and check torque per manufacturers recommendations.	Annually	N	OCS-023

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
Rail Joint Bonds	• Check the rail joint bonds for mechanical damage and tight fit of the cable lugs.	Annually	N	OCS-024
Tensorex C+ Visual Inspection	• Tensorex C+ according to manufacturer do NOT require any maintenance procedure, or operations. Ropes and their fittings are stainless steel and do not require to be greased. Bearings are pre-greased and sealed. All the components are corrosion free. • Nevertheless, Tensorex C+ require a visual inspection and possible adjustments. • *In case of newly installed lines/ conductors or refurbished copper, check the setting monthly until conductor's lengths are asserted/ stabilized. • *In the case of existing lines/ conductors check the settings at minimum 1 or 2 times per year (Annually or Semi-Annually) depending on traffic frequency, which is from consumption of the copper contact wire. Hence, typical recommended review Annually, but judgement	Semi-Annual *Annually	N	OCS-025

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
	is necessary according to individual cases for increased inspection cycles. • Checking of Tensorex C+ functionality. 1. It is possible to verify that Tensorex C+ works correctly by checking the midpoint anchor and the pointer position. o Midpoint anchor must be geometrically symmetrical. It means that both tensioning devices are applying the same pull force and work properly. o The setting scale indicates the movement of the contact line. By checking the pointer position in different temperatures, you know if Tensorex C+ is free and works properly. 2. Front Bolts. o Check the correct mounting of the bolts on the triangular front balance (yoke plate), which is			

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
	connected to the contact or messenger line, respectively. 3. Checking of the setting and recommissioning. o 1. Check whether the Tensorex C+ pointer is on the correct setting number. For information about how to calculate it see corresponding Temperature table pertaining to respective unit from manufacturer, Mosdorfer. Calculate the setting number by crossing the ambient temperature and the tension length. o 2. As an example, correct/ calculated setting number is 375 mm from temperature table. - If the actual pointer checked on site is on 330 mm, then (correct) 375 – (actual) 330 = +45. Adjust compensating fittings +45 mm longer. - If the actual checked on site is on 400 mm, then (correct) 375 – (actual) 400 = - 45. Adjust the compensating fittings - 45 mm shorter.			

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
	○ 3. Install a hoist and unload the compensating fittings to adjust. ○ 4. Adjust the turnbuckle or other compensating fittings to increase or reduce the length in accordance with the calculations. ○ 5. Release the hoist and check the pointer position again.			
Track-to-Track Cross Bonds	• Check the track-to-track cross bonds for mechanical damage and tight fit of the cable lugs. Recommend thermographic survey under load with any hot spots recorded.	Annually	N	OCS-026
Surge Arresters (Visual)	• Surge arresters recommended visual inspection quarterly. • Any locations showing obvious failed or blown arresters logged for immediate replacement, as early as possible, ahead of typical hands-on annual inspection.	Quarterly	Y – (see below Surge Arresters Hands-on)	OCS-027

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
Surge Arresters (Hands-on)	• Surge arresters should be periodically tested against leakage, 5-year cycle is recommended. • Inspect all surge arresters for failures and tracking or any damage. • Clean surge arresters and inspect all connections. Check torque on all attachments, fasteners, and stud nuts per manufacturers recommendations. • Inspect cables with split bolt fittings installed for tightness and insulation protection. • Inspect all compression fittings for cracking and torque per manufacturers recommendations. • If arresters look defective, replace, and perform testing in shop for failure or leakage, if feasible, or send out to independent testing laboratory. • Note all split bolt connections and document for change.	Annually	Y – Monitor and record locations with surge arresters' issues, failures, and replacements. Document surge arrester model, location, any available details during failure, and save failed arresters for further examination.	OCS-028

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
Up Close Inspection of OCS	• Check the adjustment of the wires in overlap sections and at crossovers. • Check the contact and messenger wires, head-span wires, hangers (dropper) wires and related thimbles, ground wires, and feeder wires for corrosion, damage, broken strands, and evidence of arcing. Wires should be monitored for arcing due to hard spots, particularly near clamps or supports. • Catenaries: o Verify the position and condition of hangers, cantilevers, steady arms, clips, and other attachments. o Verify cantilever lateral movement according to design and verify proper clearance envelope.	Annually	Y – Monitor and record locations for required adjustments and log repeat areas of OCS issues.	OCS-029
Contact Wire Hands-on Inspection	• Check the contact wire for twists, kinks and (“hard”) spots of arcing that have not previously been identified. • Check the smooth running of the contact wire with a test pantograph or level for special attention areas such			OCS-030

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
	as crossovers, section insulators, or where recent repairs have been performed. • Check CW Tensions using a ratchet lever hoist by pulling “a bubble” with dynamometer to record tension (in pounds). Both CW and MW tensions are to be checked and adjusted accordingly to maintain plumb hangers and a balanced OCS system. • Fixed Termination locations are of key concern. Refer to appropriate OCS Technical Sheets for Contact Wire/ Tension Charts, record wire temperature and approximate equivalent span length. Make necessary adjustments to fixed terminations and turnbuckles. • Recommend initially check CW tension within six months of start of revenue service on new alignments (Semi-Annual), then continue to perform Annually.	Initial inspection Semi-Annual, then Annually	Y – Monitor and record locations for required adjustments and log repeat areas of OCS issues.	

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
Messenger Wire	• Check Messenger wire for any damages. This includes damaged strands, twists, kinks such as from previous clamps or supports locations. Review any bird-caging at MW or related Jumpers. • Verify proper plumbness, or lateral displacement, tolerance is 1-inch, between MW and CW below at supports. Adjust as necessary refencing OCS As-Builts/Redlines Height & Staggers measurements. • Check MW Tensions using a rachet lever hoist by pulling “a bubble” with dynamometer to record tension (in pounds). Both CW and MW tensions are to be checked and adjusted accordingly to maintain plumb hangers and a balanced OCS system. • Fixed Termination locations are of key concern. Refer to appropriate OCS Technical Sheets for Messenger Wire/ Tension Charts, record wire temperature and approximate equivalent span length. Make necessary adjustments to fixed terminations and turnbuckles.	Initial inspection Semi-Annual, then Annually	N	OCS-031

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
	• Recommend initially check CW tension within six months of start of revenue service on new alignments (Semi-Annual), then continue to perform Annually).			
Midpoint Anchor	• Check that midpoint cantilever is perpendicular to the track and properly fixed in this position. If required, adjust length of midpoint anchor wires accordingly. • Check the symmetrical sag of both the midpoint guy wires. It means that both tensioning devices are applying the same pull force and work properly. • Check the clearance from the running level of the pantograph to the guy wire anchor. • Check guy wire tension using calibrated cable tension meter. Record tension and conductor temperature. Refer to appropriate MPA Tension Charts. Make necessary adjustments to turnbuckles and re-confirm tension.	Annually	N	OCS-032

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
Pole Mounted Disconnect Switches	• Check contact blades for dirt and arcing damage. Connecting blades should be free of cracks and arcing damage. • Check all moving parts of the switch assembly, operating link, contact blades, and operating handle for free movement. Adjust, as necessary. Lubricate according to OEM guidelines.	Annually	N	OCS-033
Balance Weight Assembly (BWA)	• Check for free movement of the pulley wheel. • Check position of weight stack depending on temperature. • Check alignment of wires on the pulley. • Adjust and lubricate pulleys/ cables, as necessary. • Check that appropriate fittings are greased according to manufacturer's specifications. • During summer months perform visual inspection once the ambient temperature reaches eighty-five degrees Fahrenheit (forecasted to reach 85F). Observation for	Annually	N	OCS-034

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
	BWA bottoming out which will require scheduling outage to adjust accordingly.			
Cables	• Inspect the cable support and termination for damage or loose connections.	Annually	N	OCS-035
OCS Structure Free of Trees and Bushes	• Visually check that the OCS and conductors are a minimum 10-feet from trees, shrubs, bushed, and any vegetation. • The absolute minimum electrical clearance for energized OCS-to-Ground of High concern is 5-inches per Sound Transit Specifications Section 34 23 01 for OCS General Requirements. Trim any trees, vegetation back to 10-feet from OCS. • Record any locations of high concern to be addressed accordingly and ahead of Annual PM, if necessary.	Quarterly	Y – Monitor and record locations for repeat occurrences to perform further tree cutting, pruning or removal, as necessary.	OCS-036
Steady Arm	• Check for any excessive steady arm along track movement with respect to main assembly.	Annually	N	OCS-037

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
Foundations	• Condition of concrete, visible cracks, deposits of rubbish, overgrowth of vegetation and concrete condition.	Quarterly	Y – Monitor and record locations for repair or replacement with worsening conditions, adjust inspection cycle if necessary.	OCS-038
Poles	• Check poles for loose hardware on bonding cables, broken or cracked welds, damaged galvanization, distortion, cracking, or corrosion. • Check the completeness, cleanliness, and proper attachment of signage and pole numbers.	Quarterly	Y – Monitor and record locations for repair or replacement with worsening conditions, adjust inspection cycle if necessary.	OCS-039
Cables	• Check the integrity of grounding and bonding cables. • Check cable terminations for damage or dirt.	Quarterly	N	OCS-040
Contact Wire	• Visual inspection for wire damages at contact wire supports.	Quarterly	N	OCS-041

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
	• Review the position of contact wire bridges at crossovers and the position of wires at overlap and crossovers.			
Catenaries	• Visual inspection of catenary for broken or displaced hangers. Repair/Replace accordingly and ahead of Annual PM, if necessary.	Quarterly	N	OCS-042
I-90 Floating Bridge Catenary Monitoring System (CMS)	• Check CMS HMI Screen for "Station OK" and "Sensor OK." Ensure no alarms/indications present for "Cable Impact", "Cable Rupture", "Sensor Cable Break", "Sensor Magnet Lost", "FO Fault", and "Station Fault."	Quarterly	N	OCS-043
I-90 Tension Wheel Assembly	• Check whether external influences can hinder movement of the wheel body. • Check the steel wires for damage e.g. wire breaks, flattened surfaces, and wear. • Check the connections to the steel wires.	Annually	N	OCS-044

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
	• Check all mechanical connections resp. the fixing elements. • Check for correct positioning according to wire temperature and wire run length.			
I-90 Balance Weight Assembly	• Check the balance weight assembly for corrosion of steel wire. Adjust, as necessary. • Check for correct positioning according to wire temperature and wire run length.	Annually	N	OCS-045
I-90 Portal Structures (All types)	• Check integrity of assembly. Check for loose hardware, broken or cracked welds, damaged galvanization, distortion, cracking, or corrosion.	Quarterly	N	OCS-046
I-90 General	• Check position of insulators, steady arms and contact wire clips. Adjust, as necessary. • Check OCS switches and cantilever assemblies for slackened or missing jumpers.	Quarterly	N	OCS-047
I-90 Cables	• Inspect exposed sections of cables for physical damage or evidence of overheating.	Quarterly	N	OCS-048

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
	• Inspect messenger wire for abrasions and bird-caging. • Inspect terminations, jumpers, and splices for evidence of physical damage or overheating. • Inspect for signs of OCS hard spots			
I-90 Floating Bridge Stray Current and Corrosion Control Remote Monitoring System (SCCCRMS)	• Track Electrical Isolation Monitoring System: o Conduct Quarterly Track-to-earth resistance testing on SCCCRMS. Minimum acceptance criteria to be 1250 ohms-1000 feet. Adjust frequency of testing according to data collected and trending observed, increase to monthly or weekly, as necessary. • Stray Current Collector: o Visually inspect cables for damage or cut insulation. Inspect connections are intact. o Clear any debris that may cause electrical bridge from rails to bridge deck.	Quarterly	Y – Monitor and record locations readings and changes. Adjust inspection cycle, as necessary.	OCS-049

Element	Activity	Frequency Required Annually Semi-Annual Quarterly	Single Point Data Consolidation and Trending Required (Y/ N)	Task Reference Number
	○ Conduct collector mat continuity test on SCCCRMS. • Pontoon Cathodic Protection: ○ Check pontoon cathodic protection. Ensure anchor cable potential, cable coupon potential, structure potential and structure coupon potentials do not exceed - 0.85V. ○ Check electrical continuity of pontoons and anchor cables.			

Table Preventative Maintenance Task Table-6 OCS Maintenance Task Table

8.2 OCR Maintenance Task Table

Table Preventative Maintenance Task Table- 7 OCR Maintenance Task Table

Element	Activity	Frequency Required Annually Quarterly	Single Point Data Consolidation and Trending Required	Task Reference Number
Rigid Supports	• Inspect the torque on bolts per manufacturers recommendations. • Inspect the condition of bolts and galvanized surfaces. • Inspect the adjustment of heights, cants, and staggers. • Replace bolts if they are corroded. • Protect or replace any exposed galvanized parts, replace if the corrosion is excessive.	Annually	N	OCR-001
Parafil or Steel-wire Head Spans	• Inspect the torque on bolts per manufacturers recommendations and mechanical tension on wires. • Clean Parafil synthetic rope or steel-wire cables with a cloth and alcohol. • Inspect adjustment of heights and staggers. • Replace bolts if they are corroded. • Replace any damaged part.	Annually	N	OCR-002
Bell Insulator with sliding	• Inspect the torque on bolts per manufacturers and adjust accordingly. The Allen bolts of the hanger clamp	Annually	N	OCR-003

Element	Activity	Frequency Required Annually Quarterly	Single Point Data Consolidation and Trending Required	Task Reference Number
hanger clamp	have a special torque of 22Nm (16.6 lb/ft). • Check the condition of the polyamide pads inside the hanger clamp (this allows the catenary bars to freely slide inside the hanger clamp). • Check that the aluminum bar can slide inside the hanger clamp. • Clean the surface of the insulator with a cloth and alcohol. • Replace the insulated hanger assembly if the insulator is broken or the polyamide pads are damaged.			
OCR bar	• Inspect for corrosion marks. • Inspect for presence of debris. • Inspect that the contact wire is well inserted into the aluminum bar. • If there are corrosion marks: Black marks: Clean with a metal brush. Aggressive corrosion: Stop or divert the water filtration, replace the bar, and install a protective cover.	Annually	N	OCR-004

Element	Activity	Frequency Required Annually Quarterly	Single Point Data Consolidation and Trending Required	Task Reference Number
Contact Wire	• Check the condition of the contact wire. • If the wear on the contact wire is more than 50% of its original section (or 30% in transitions with mechanical tension), it is necessary to replace the contact wire. If there are burns, check the leveling and the electrical circuit. Clean and file any burn marks and pitting damage.	Annually	Y – Monitor and record locations readings and changes. Adjust inspection cycle, as necessary.	OCR-005
Splice Joints of Aluminum profile	• Check the torque on bolts per manufactures recommendations (22 Nm, 16.6 lb/ft). • Check that both kinds of washers are in place (flat and spring). • Check there are no corrosion marks.	Annually	N	OCR-006
Transition Bar	• Check the torque on bolts (60 Nm, 44lb/ft). • Check the placement of the contact wire inside the bar. • Check the position of the contact wire splice inside the transition bar slot (if the transition bar has this equipment). • Clean the protective cover.	Annually	N	OCR-007

Element	Activity	Frequency Required Annually Quarterly	Single Point Data Consolidation and Trending Required	Task Reference Number
Mid-point/ transition anchoring	• Check the general condition of the mid-point clamp and tighten the Parafil synthetic ropes (so they are in tension, but without raising the aluminum bar excessively). • Check that the mid-point clamp is fully clamped into the profile and has not slipped. Verify the torque on the anchoring clamp. • Check the torque on bolts: 30 Nm, (22 lb/ft) for mid-point anchoring, or 40 Nm, (29 lb/ft) for transition anchoring. • Clean Parafil synthetic ropes with a cloth and alcohol.	Annually	N	OCR-008
Power Feed Clamp	• Check that the clamp is fully inserted into the profile. • Check the torque on the threaded rod (60 Nm, or 44 lb/ft). Check the torque on the bolt for lugs (43 lb/ft). • Check the status of the electrical connections.	Annually	N	OCR-009
Anchor Clamps Set	• Check that anchor clamps are fully inserted into the profile and have not slipped. • Check the torque on bolts (30 Nm, or 22 lb/ft).	Annually	N	OCR-010

Element	Activity	Frequency Required Annually Quarterly	Single Point Data Consolidation and Trending Required	Task Reference Number
Grounding Clamp	• Check that the clamp is fully inserted into the profile. • Check the torque of the threaded rod 60 Nm, (44 lb/ft). • Inspect the status of the stirrup.	Annually	N	OCR-011
OCR Expansion Joint (CR-EXP)	• Steel structural components and support: o Check torque markings and/or actual torque on bolts per manufacturers recommendations. o Check condition of bolts (galvanized/finished) surfaces. • Standoff insulators: o Check torque markings and/or actual torque on bolts per manufacturers recommendations. o Check condition of bolts (galvanized/finished) surfaces. o Check condition of insulator surface for contamination, damage, or crack. • Termination and Roller Support Assembly: o Check torque markings and/or actual torque on bolts. o Check condition of bolts (galvanized/finished)	Annually	N	OCR-012

Element	Activity	Frequency Required Annually Quarterly	Single Point Data Consolidation and Trending Required	Task Reference Number
	surfaces. o Check air gap clearance at contact wire. o Check condition of rollers. o Check condition of pin and sleeve bearing connection. • OCR (Transition/Ramp) bars and Splice Joints: o Check torque on bolts and all other hardware condition. o Check bar surface condition or any corrosion. o Check for any accumulated dirt or debris. o Check the position of splice joints. o Check alignment and contact wire alignment on its profile. o Check protective cover.			
OCR Jumper Assembly (CR-XJ)	• Cable Support Assembly and Cleat: o Check torque markings and/or actual torque on bolts. o Check condition of bolts (galvanized/finished)	Annually	N	OCR-013

Element	Activity	Frequency Required Annually Quarterly	Single Point Data Consolidation and Trending Required	Task Reference Number
	surfaces. o Check condition of cleat for damage or deterioration. o Check that jumper assembly does not sag below in-running OCR. • Power Feed Block and Lugs: o Check torque markings and/or actual torque on bolts. o Check condition of bolts (galvanized/finished) surfaces. o Check condition of bimetallic plate. o Check condition of compression lug.			
OCR Anchor Bracket (CR-AB)	• Steel structural components and support o Check torque markings and/or actual torque on bolts. o Check condition of bolts (galvanized/finished) surfaces. o Check caps and covers.	Annually	N	OCR-014

Element	Activity	Frequency Required Annually Quarterly	Single Point Data Consolidation and Trending Required	Task Reference Number
	• Bar and Standoff insulators o Check torque markings and/or actual torque on bolts. o Check condition of bolts (galvanized/finished) surfaces. o Check condition of insulator surface for contamination, damage, or crack insulator. • OCR (Transition/Ramp) bars o Check torque on bolts and all other hardware condition. o Check bar surface condition or any corrosion. o Check for any accumulated dirt or debris. o Check alignment and contact wire alignment on its profile. • Power Feed Block and Lugs o Check torque markings and/or actual torque on bolts. o Check condition of bolts (galvanized/finished) surfaces.			

Element	Activity	Frequency Required Annually Quarterly	Single Point Data Consolidation and Trending Required	Task Reference Number
	o Check condition of bimetallic plate. o Check condition of compression lug.			
OCR Expansion Joint (CR-EXP)	• Standoff insulators: o Check condition of insulator surface for contamination, damage, or crack. • Termination and Roller Support Assembly: o Check air gap clearance at contact wire. o Check condition of rollers. o Check condition of pin and sleeve bearing connection. • OCR (Transition/Ramp) bars and Splice Joints: o Check bar surface condition or any corrosion. o Check for any accumulated dirt or debris. o Check the position of splice joints. o Check alignment and contact wire alignment on its profile. o Check for smooth pantograph transition. o Check protective cover.	Quarterly		OCR-015

Element	Activity	Frequency Required Annually Quarterly	Single Point Data Consolidation and Trending Required	Task Reference Number
Quarterly	• Cable Support Assembly and Cleat: o Visually review condition of cleat for damage or deterioration. o Visually review that jumper assembly does not sag below in-running OCR. • Power Feed Block and Lugs: o Check condition of bimetallic plate. o Check condition of compression lug.	Quarterly		OCR-016

9.0 REFERENCES

9.1 IEEE Std 1791-2019 IEEE Recommended Practice for Terminology Used for Direct Current Electric Transit Overhead Contact Systems

9.2 APTA-RT-FS-S-006-03-Rev 2 – 2017 Traction Electrification Distribution System Inspection, Maintenance and Testing

9.3 IGA 2019 KCC Ordinance

9.4 Sound Transit Requirements Manual (ST RM) Rev 1 – DEC 2024

10.0 APPENDICES

10.1 Appendix A – Asset Hierarchy

10.2 Appendix B – Torque Specification Tables

10.2.1 US Standard Fasteners – Heat-Treated Steel Cadmium or Zinc Plated

10.2.2 US Standard Fasteners – Silicon Bronze

10.2.3 US Standard Fasteners – Aluminum Alloy

10.2.4 US Standard Fasteners - Stainless Steel

10.3 Appendix C – OCS Maintenance Standard RACI (Responsible, Accountable, Consulted, Informed) Matrix

10.4 Appendix D – OCS Acceptance Measurements (per ST Specs 34 23 69 & ST RM)

10.5 Appendix E – Reference Sample OCS Test Procedures attached with this standard

10.6 Appendix F - Tensorex C+ Spiral Spring Tensioning Device

10.6 Appendix F – N830 E750 NG EL MW & CW Temperature/Tension Charts

10.7 Appendix G – L800 LLE MW & CW Temperature/Tension Charts

10.8 Appendix H – F200 MW & CW Temperature/Tension Charts

10.9 Appendix I – U830 University Link Extension MW and CW Temperature/Tensions Charts

10.10 Appendix J – Figure 1 – 350 MCM Contact Wire Wear

10.11 Appendix K – Figure 2 – Tensorex C+ Spiral Spring Tensioning Device (by Mosdorfer) for Auto- Tensioned (AT) OCS

10.1 Appendix A – Asset Hierarchy

Sound Transit's assets are critical components throughout its systems. Refer to the latest version of OCS Asset Plan to see asset hierarchy table as work is under way with Asset Management group. Sample of Assets, below.

System	Sub-System	Asset	Component/Parts	Component/Part	Design Life Age
Overhead Contact System					
	OCS	OCS Tapered Tubular Pole			
			Concrete		
			Vertical Rebar		
			Horizontal Rebar		
			Anchor Bolt		
			Anchor Bolt Nut		
			Anchor Bolt Washer		
			Anchor Plate		
			Bolt Pattern Template		
			Ground Rod		
			Copper Cable		
			Ground Connector		
			Feeder Conduit		
		OCS Down Guy Anchor			
			Concrete		
			Vertical Rebar		

System	Sub-System	Asset	Component/Parts	Component/Part	Design Life Age
			Horizontal Rebar		
			Anchor Rod		
			Copper Cable		
			Ground Connection		
		OCS Wide Flange Pole			
			Concrete		
			Vertical Rebar		
			Horizontal Rebar		
			Anchor Bolt		
			Anchor Bolt Nut		
			Anchor Bolt Washer		
			Anchor Plate		
			Bolt Pattern Template		
			Ground Rod		
			Copper Cable		
			Ground Connector		
	I-90 Floating Bridge Systems	OCS Portal Assembly			
			Standard Portal		
			Midpoint Portal		
			Balance Weight Portal		
			Fixed Terminated Portal		
			Mixed Portal		
			4/0 Insulated Grounding Cable		
			Ground Mat		
			Aerial Grounding Cable		

System	Sub-System	Asset	Component/Parts	Component/Part	Design Life Age
		Portal Hinge Base			
			Hinge Plate		
			Hinge Pin and (2) Collars		
			1/4" Thick Insulation		
			Portal Mounting Nuts		
			Portal Mounting Jam Nuts		
			Portal Mounting Washers		
			Portal Mounting Bolt Insulation		
			1/4" Thick Stainless-Steel Bushing		
			1/4" Thick Steel Plate		
	OCR	Conductor Rail Bridge Expansion Assembly			
			Vertical Tube		
			Beam Brackets for Insulators		
			Stand Off Insulators		
			Conductor Rail Rigid Bracket		
			Conductor Rail Glider Bracket		
			W-section W/ Connection		
			Transition Section		
			End Section		
			Splice Clamp		
			Conductor Rail		
			Plastic Cover		
			Diagonal Brace		
			Beam Brackets for Wide Flange-section		

System	Sub-System	Asset	Component/Parts	Component/Part	Design Life Age
		Pole			
			Foundation		
			Tubular Pole		
			Insulators		
			Cantilever		
			Contact Wire		
			Messenger Wire		
			Jumper Wire		
			Pole Mounted Switches		
			Hangers		
			Section Insulators.		
			In-Span Insulators		
	OCS		Surge Arresters		
			Down Guy		
			Head Guy		
			Headspan		
			Mid-Point Anchor		
			Mid-Point		
			Brackets		
			Pull-Off		
			Clamp		
			Pole Bands		
			Spring Tensioner		
			Termination		
			Feeder Cables		

System	Sub-System	Asset	Component/Parts	Component/Part	Design Life Age
			Fixed Anchors.		
			Y-Terminations.		
			Constant-Tension Spring Terminations.		
			Knuckles.		
			Contact Bridges.		
			Splices.		
			Bridles.		
			Anchor Brackets.		
			OCS Monitoring System.		
			Cross Span		
			Tunnel Support?		
			Registration		
		Balance Weight Assembly			
			Clamps		
			Thimble		
			Guide Pipe		
			Insulator		
			Dead-End		
			Shackle		
			Wire Rope		
			Turnbuckle		
			Eyebolt		
			Sleeve		
			Wheel		

System	Sub-System	Asset	Component/Parts	Component/Part	Design Life Age
			Bracket		
			Slug		
	OCR	Tunnel OCR			
			Rigid Supports		
			Steel Wire Head Span		
			Bell Insulator with sliding hanger clamp		
			Aluminum Profile		
			Contact Wire		
			Splice Joints of Aluminum Profile		
			Transition Bar		
			Mid-Point/Transition Anchoring		
			Feeder Clamp		
			Anchor Clamps Set		
			Earthing Clamp		
	I-90 Floating Bridge Systems	Tension Wheel Assembly/Catenary Monitoring System			
			Wheel Body		
			Swing Lever		
			Suspension Latching Plate		
			Guide Tube Bracket		
			Safety Hoop		
			Sensor Rod		
			Permanent Magnet		
			Sensor Fastenings		

System	Sub-System	Asset	Component/Parts	Component/Part	Design Life Age
			Sensor Cable		
		Expansion Joint Assembly (CR-EXP)			
			Beam Steel Structure		
			Column Steel Structure		
			Insulated Support Assembly		
			OCR Support Column		
			Roller Support		
			Termination Assembly		
			Splice Plate		
			Contact Wire Clamp		
			Sensor Cable		
		Jumper Assembly (CR-XJ)			
			Cable Support Assembly		
			Power Feed Block		
			Compression Lug		
			Cable Cleat		
		Anchor Bracket Assembly (CR-AB)			
			Vertical Support Weldment		
			Support Clamp Weldment		
			Stiffening Weldment		
			Track Support Weldment		
			Support Bracket for Bar Insulator		
			Bar Insulator		

System	Sub-System	Asset	Component/Parts	Component/Part	Design Life Age
			Anchor Clamp		
			Power Feed Block		
		Stray Current and Corrosion Control Remote Monitoring System (SCCCRMS)			
			Track Electrical Isolation Monitoring System		
			Stray Current Collector		
			Cathodic Protection System		
			Collector Cables		
			Pontoon Coupon		

10.2 Appendix B – Torque Specification Tables

10.2.1 US standard Fasteners - Heat-Treated Steel Cadmium or Zinc Plated

Table Appendices-3 US standard Fasteners - Heat-Treated Steel Cadmium or Zinc Plated

Grade	SAE 1&2	SAE 5	SAE 7	SAE 8
Minimum Tensile (P.S.I.)	64K	105K	133K	150K
Bolt Diameter (In Inches)	Torque (Foot Pounds)			
¼”	4.0	5.6	8.0	8.4
5/16”	7.2	11.2	15.2	17.6
3/8”	12.0	20.0	27.2	29.6
7/16”	19.2	32.0	44.0	48.0
½”	29.6	48.0	68.0	73.6
9/16”	42.4	70.4	96.0	105.6
5/8”	59.2	96.0	133.6	144.0
¾”	96.0	160.0	224.0	236.8
7/8”	152.0	241.6	352.0	378.4
1.0”	225.6	372.8	528.0	571.2

10.2.2 US Standard Fasteners – Silicon Bronze

Table Appendices-8 US Standard Fasteners- Silicon Bronze

Bolt Diameter (in Inches)	Non-Lubricated (Foot-Pounds)	Lubricated (Foot-Pounds)
5/16”	15	10

Bolt Diameter (in Inches)	Non-Lubricated (Foot-Pounds)	Lubricated (Foot-Pounds)
3/8”	20	15
½”	40	25
5/8”	55	40
¾”	70	60

10.2.3 US Standard Fasteners – Aluminum Alloy

Table Appendices- 9 Table Appendix A-3: US Standard Fasteners- Aluminum Alloy

Bolt Diameter (in Inches)	Non-Lubricated (Foot-Pounds)	Lubricated (Foot-Pounds)
5/16”	13	8
3/8”	16	11
½”	35	20
5/8”	47	32
¾”	60	48

10.2.4 US Standard Fasteners – Stainless Steel

Table Appendices- 10 US Standard Fasteners: Stainless Steel

Bolt Diameter (in Inches)	Non-Lubricated (Foot-Pounds)	Lubricated (Foot-Pounds)
5/16”	14	10
3/8”	25	20

Bolt Diameter (in Inches)	Non-Lubricated (Foot-Pounds)	Lubricated (Foot-Pounds)
½”	45	30
5/8”	60	45
¾”	90	80

10.3 Appendix C – OCS Maintenance Standard RACI (Responsible, Accountable, Consulted, Informed) Matrix

OCS Maintenance Standard RACI Matrix will be included in a future revision.

10.4 Appendix D – OCS Acceptance Measurements (per ST Standard Specifications 34 23 69 and ST Requirements Manual)

1. OBJECTIVE:

The purpose of these measurements is to ensure the OCS splice is installed as per the existing measurements and within the specified tolerances.

2. Test Prerequisites:

Prior to taking these measurements, verify that the OCS splice and terminations are installed in accordance with the specifications and is inoperable condition. Make sure it is verified through the OCS Visual inspection.

3. Personnel Required:

KCM: for this activity KCM crew will be taking measurements and recording on the test sheet.

Quality representative: Sound Transit SMEs will be onsite to witness the measurements.

4. Test Equipment

This test procedure requires:

- Insulated telescoping height measuring stick and/or a laser measuring device.
- Stagger gauge and/or laser measuring device
- Hi-Rail Platform Truck
- Hi-rail Bucket Truck or other equipment
- All calibrated equipment used shall be within its active calibration interval
- Equipment Calibration Certifications provided with Test Report

5. Test Method and procedure:

Upon completion of all construction and completion of adjustments resulting from the visual inspection, the following items will be measured using equipment notes above to ensure compliance with specifications and recorded on the OCS Acceptance Measurement Data Sheet.

- Contact Wire Height
- Messenger wire height

- Stagger - Distance from contact wire to super-elevated centerline of track measured at support location.
- Measure stagger to nearest 1/2 inch.
- Record as Left (L) or Right (R) when viewed in direction of increasing stationing.
- Cross Level (Superelevation).
- In-Running Contact Wire Height: Height of in-running contact wire at overlaps and turnouts.
- Out-of-Running Contact Wire Height: Height of out-of-running contact wire at overlaps and turnouts.
- Compute the following Using Measurements Taken:
- Sag:
    - Compute halfway between structures.
    - Average of contact wire height for previous and next point of attachment minus midspan contact wire height.

6. PASS/ Fail Criteria

Pass Criteria

The acceptance measurements deemed Pass if all measurements are within construction installation tolerances per projects Specifications Section 34 23 26 3.01K as follows:

- The installed OCS shall conform to the dimensional requirements shown on the Contract Drawings within the following tolerances:
    - Contact wire height in tunnels: - 0 inch, +1/2 inch
    - Contact wire height in open route: - 0 inch, + 1 inch
    - Contact wire height in stations: - 0-inch, +1 inch
    - Contact wire stagger at registration: +/- 1/2 inch
    - System height: 1 inch
    - Hanger location: 3 inches
    - Hanger length: 1/4 inch

- Messenger to Contact Wire lateral displacement at support (plumbness): 1 inch
- Pole base elevation: 1 inch
- Pole rake (after loading): 1 inch (at top)
- Wire tension: +/- 3% based on the CTST nominal load.

Fail Criteria

The acceptance measurements deemed Fail at locations where measurements are not within the above construction installation tolerances. At these locations, KCM will make the necessary adjustments, and the new measurements will be recorded.

(**See sample OCS Acceptance Measurements Data Sheet on next page**)

ST- OCS Acceptance Measurements Data Sheet

Date:_____ Wire Run:_____ Track_____ Drawing Number:_____

Ambient Temprature:_____ Contact Wire Temprature:_____ OCS Type:_____ Measuring Equipment:_____

Structure									Mid Span(Taken between each assembly					Sag & Gradient		Overlaps & Turnouts		Terminations				Wire Tenstion			
				CW Height (ft.-in)		MW Height (ft.-in.)					Contact wire Height (ft.-in.)					Stagger CW Height		CW Termin. Height (ft.-in)		MW Termin. Height (ft.-in)					
Structure #	Structure Station	Super elevation (in)	Stagger (L or R) Nearest 1/2in	FT	IN	FT	IN	Span Length (ft.-in.)	Offse/Stagger (in.)	Span Length (ft.)	FT	IN	Super-elevation (in.)	Pre. Sag (in.)	Gradient (ft/ft)	Overlap (L or R)	Overlap (ft.-in.)	FT	IN	FT	IN	CW Guage Reading	MW Guage Reading	Turnbuckle Status	Remarks

Measurements By: _____ Date: _____
QC Witness: _____ Date: _____
Sound Transit Witness: _____ Date: _____

10.5 Appendix E – Reference Sample OCS Test Procedures attached with this standard.

10.5.1 N830 E750-REC-04216 SUB 342369-014.002 1.03A.1.b OCS Acceptance Measurements Test Procedure

10.5.2 N830 E750-REC-04215 SUB 342369-008.002 1.03A.1.b OCS Visual Inspection Test Procedure Test Procedure

10.5.3 N830 E750-REC-04274 SUB 342369-012.002 1.03A.1.b OCS Electrical and Mechanical Clearance Test Procedure

10.5.4 N830 E750-REC-04276 SUB 342369-007.002 1.03A.1.b OCS Continuity Test Procedure

10.5.5 N830 E750-REC-04277 SUB 342369-004.002 1.03A.1.b OCS Hi-Pot Dielectric Test Procedure

Note: Per ST Requirement Manual, clause 221.6.7.6: DC Hi-Pot insulation testing must use a test voltage of 4.5 kilovolts and the leakage current to ground must be less than or equal to 5 milliamperes per mile of single-track OCS.

10.5.6 N830 E750-REC-04275 SUB 342369-006.002 1.03A.1.b OCS Earth Ground Resistance Test Procedure

10.5.7 N830 E750-REC-06390 SUB 019500-048.003 1.05A SIT 301-302 OCS Live Wire – Supplemental Response

10.6 Appendix F – N830-E750 NG EL MW & CW Temperature/Tensions Charts

N830 Northgate Link Extension and E750 East Link Extension

Messenger Wire (MW) and Contact Wire (CW) Temperature/Tensions Chart (AT)

ERECTION TENSIONS FOR AUTO-TENSIONED CONTACT WIRE				
L = EQUIVALENT SPAN (FT)		22	25	30
tn = NEW TEMPERATURE	0	5937	5935	5931
	10	5497	5496	5492
	20	5057	5055	5052
	30	4617	4616	4613
	40	4177	4176	4174
	50	3738	3738	3737
	60	3300	3300	3300
	70	2862	2863	2865
	80	2427	2430	2434
	90	1995	2000	2008
	100	1571	1580	1597
	110	1165	1183	1213
	120	809	839	889

ERECTION TENSIONS FOR REDUCED AUTO-TENSIONED CONTACT WIRE				
L = EQUIVALENT SPAN (FT)		22	25	30
tn = NEW TEMPERATURE	0	5132	5127	5120
	10	4691	4688	4680
	20	4251	4249	4242
	30	3812	3810	3804
	40	3374	3372	3367
	50	2936	2935	2932
	60	2500	2500	2500
	70	2067	2069	2074
	80	1641	1647	1659
	90	1231	1245	1269
	100	863	890	933
	110	588	626	686
	120	426	467	531

.

N830 Northgate Link and E750 East Link Extension CW (Fixed Termination)

EAST LINK - ERECTION TENSIONS FOR FIXED TERMINATED CONTACT WIRE							
L = EQUIVALENT SPAN (FT)		22	25	50	65	80	100
tn = NEW TEMPERATURE	0	5937	5935	5910	5886	5856	5808
	10	5497	5496	5471	5449	5422	5377
	20	5057	5055	5034	5015	4989	4951
	30	4617	4616	4598	4581	4561	4527
	40	4177	4176	4163	4150	4135	4110
	50	3738	3738	3729	3722	3713	3699
	60	3300	3300	3300	3300	3300	3300
	70	2862	2863	2875	2886	2898	2916
	80	2427	2430	2459	2484	2513	2555
	90	1995	2000	2058	2103	2153	2222
	100	1571	1580	1681	1754	1829	1929
	110	1165	1183	1349	1452	1553	1677
	120	809	839	1079	1209	1328	1472

NORTH LINK - ERECTION TENSIONS FOR FIXED TERMINATED CONTACT WIRE							
L = EQUIVALENT SPAN (FT)		22	25	50	65	80	100
tn = NEW TEMPERATURE	0	5636	5634	5601	5571	5533	5473
	10	5195	5193	5164	5135	5101	5045
	20	4756	4754	4726	4702	4670	4620
	30	4316	4314	4291	4270	4244	4202
	40	3877	3875	3857	3841	3821	3790
	50	3438	3437	3426	3417	3405	3387
	60	3000	3000	3000	3000	3000	3000
	70	2564	2564	2581	2595	2610	2632
	80	2130	2134	2174	2206	2243	2292
	90	1703	1710	1789	1847	1908	1990
	100	1289	1303	1441	1531	1619	1730
	110	913	938	1151	1271	1381	1514
	120	621	659	931	1069	1192	1340

ERECTION TENSIONS FOR AUTO-TENSIONED MESSENGER WIRE (UNLOADED)										
L = EQUIVALENT SPAN (FT)		60	75	90	100	125	145	150	175	200
tn = NEW TEMPERATURE	0	8520	8383	8217	8092	7728	7391	7301	6827	6326
	10	7897	7764	7604	7481	7130	6809	6723	6278	5819
	20	7276	7148	6992	6876	6541	6239	6159	5750	5338
	30	6658	6534	6387	6276	5963	5684	5612	5246	4890
	40	6042	5925	5787	5685	5398	5151	5087	4772	4476
	50	5430	5323	5198	5106	4856	4643	4590	4334	4102
	60	4824	4730	4623	4545	4338	4171	4130	3936	3767
	70	4227	4152	4068	4009	3856	3738	3710	3581	3472
	80	3645	3597	3544	3508	3419	3354	3338	3269	3213
	90	3088	3075	3062	3053	3032	3017	3014	2999	2987
	100	2571	2603	2635	2655	2700	2730	2736	2767	2791
	110	2115	2198	2273	2319	2420	2487	2501	2567	2622
	120	1741	1867	1978	2045	2188	2281	2303	2397	2473

N830 Northgate Link and E750 East Link Extension MW (Fixed Termination)

EAST LINK - ERECTION TENSIONS FOR FIXED TERMINATED MESSENGER WIRE (UNLOADED)								
L = EQUIVALENT SPAN (FT)		45	50	60	65	80	90	145
tn = NEW TEMPERATURE	0	8627	8595	8520	8477	8330	8217	7391
	10	8002	7970	7897	7856	7713	7604	6809
	20	7379	7347	7276	7236	7099	6992	6239
	30	6756	6726	6658	6619	6487	6387	5684
	40	6133	6105	6042	6005	5882	5787	5151
	50	5514	5489	5430	5396	5283	5198	4643
	60	4899	4876	4824	4794	4696	4623	4171
	70	4289	4269	4227	4204	4126	4068	3738
	80	3686	3673	3645	3629	3579	3544	3354
	90	3099	3096	3088	3083	3071	3062	3017
	100	2540	2551	2571	2582	2614	2635	2730
	110	2031	2059	2115	2143	2224	2273	2487
	120	1601	1649	1741	1785	1907	1978	2281

NORTH LINK - ERECTION TENSIONS FOR FIXED TERMINATED MESSENGER WIRE (UNLOADED)								
L = EQUIVALENT SPAN (FT)		45	50	60	65	80	90	145
tn = NEW TEMPERATURE	0	9153	9126	9065	9030	8910	8815	8130
	10	8527	8501	8441	8407	8290	8199	7536
	20	7902	7878	7819	7786	7673	7585	6951
	30	7279	7255	7198	7166	7058	6975	6378
	40	6656	6633	6580	6550	6447	6369	5820
	50	6035	6013	5965	5936	5841	5770	5280
	60	5416	5397	5352	5328	5244	5181	4766
	70	4800	4784	4748	4726	4658	4606	4284
	80	4191	4179	4153	4137	4089	4053	3842
	90	3592	3586	3573	3567	3545	3529	3444
	100	3008	3012	3020	3024	3039	3049	3096
	110	2454	2472	2509	2529	2586	2623	2796
	120	1955	1991	2063	2099	2201	2264	2543

10.7 Appendix G – L800 LLE MW & CW Temperature/Tensions Charts

L800 Lynnwood LE Messenger Wire (MW) & Contact Wire (CW) Temperature/Tensions Chart (AT)

ERECTION TENSIONS FOR UNLOADED AUTO-TENSIONED MESSENGER WIRE (LB)									
AMBIENT	EQUIVALENT SPAN (FT)								
	60	75	90	100	125	145	150	175	200
0	8520	8383	8218	8092	7729	7392	7302	6828	6328
10	7898	7764	7604	7482	7131	6810	6725	6280	5821
20	7277	7148	6993	6876	6542	6239	6160	5751	5341
30	6658	6534	6387	6276	5963	5685	5613	5247	4892
40	6042	5926	5788	5686	5400	5151	5088	4773	4478
50	5430	5323	5199	5107	4856	4645	4592	4335	4103
60	4824	4731	4624	4546	4339	4172	4131	3937	3768
70	4227	4153	4069	4010	3858	3740	3712	3582	3472
80	3645	3597	3544	3508	3420	3354	3339	3270	3214
90	3088	3075	3062	3053	3033	3018	3015	3000	2988
100	2571	2604	2635	2655	2700	2730	2737	2768	2792
110	2116	2198	2273	2319	2420	2487	2502	2568	2622
120	1742	1868	1979	2045	2188	2282	2303	2397	2473

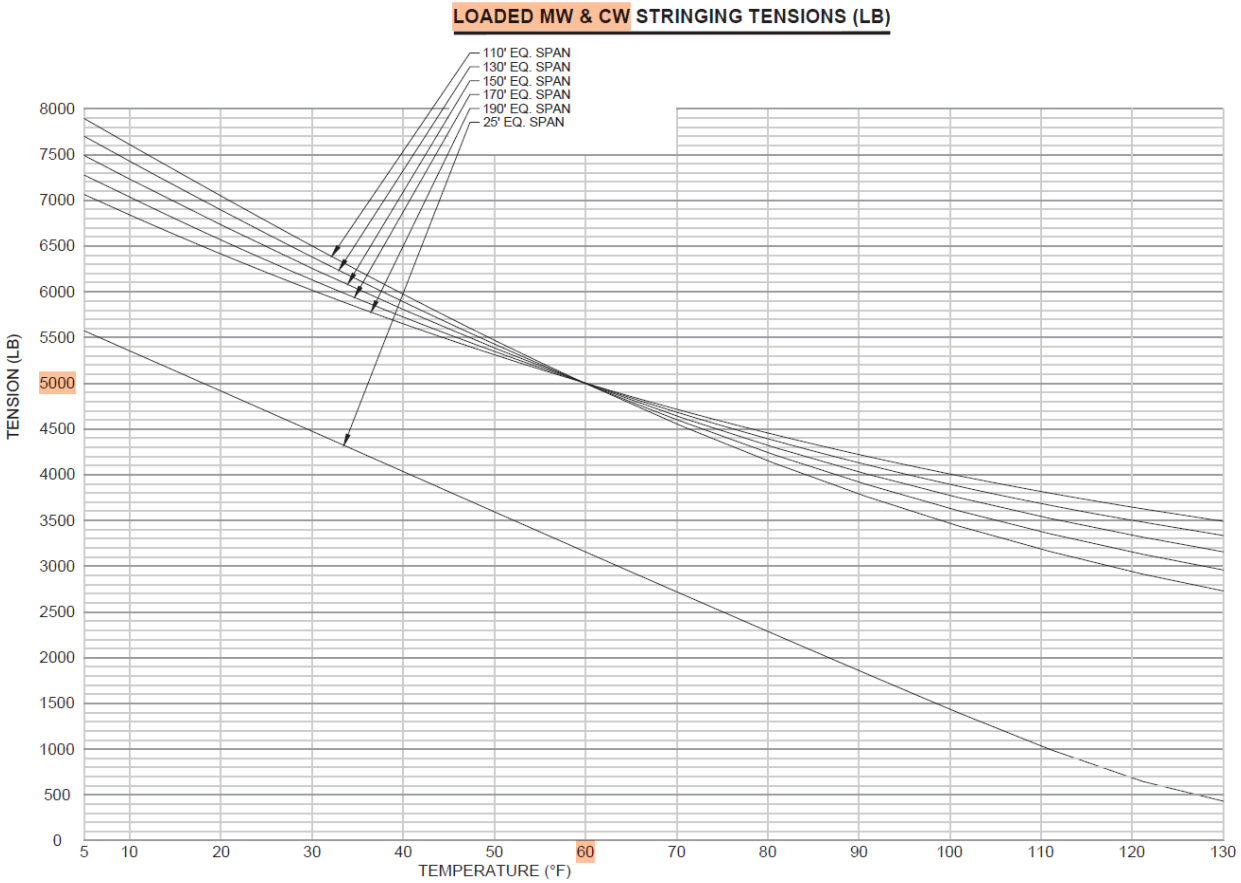
ERECTION TENSIONS FOR AUTO-TENSIONED CONTACT WIRE (LB)					
AMBIENT	EQUIVALENT SPAN (FT)				
	20	22	25	27	30
0	5939	5938	5936	5934	5932
10	5498	5497	5496	5494	5492
20	5058	5057	5056	5054	5052
30	4618	4617	4616	4615	4613
40	4178	4178	4177	4176	4175
50	3739	3739	3738	3737	3737
60	3300	3300	3300	3300	3300
70	2862	2862	2863	2864	2865
80	2425	2427	2429	2431	2434
90	1992	1995	2000	2003	2009
100	1565	1571	1580	1586	1597
110	1154	1165	1182	1194	1213
120	789	809	839	859	889

10.8 Appendix H – F200 FWLE MW and CW Temperature/Tensions Charts

F200 Federal Way LE Messenger Wire (MW) & Contact Wire (CW) Temp/Tensions Chart (Loaded) (AT)

LOADED MW & CW STRINGING TENSIONS (LB)

TEMP (°F)	EQUIVALENT SPAN (FT)					
	LOADED MW					CW
	110	130	150	170	190	25
5	7891	7695	7486	7271	7058	5716
10	7606	7423	7229	7031	6837	5496
20	7045	6891	6730	6569	6413	5056
30	6500	6378	6254	6132	6017	4616
40	5974	5890	5805	5724	5649	4177
50	5473	5429	5386	5347	5310	3738
60	5000	5000	5000	5000	5000	3300
70	4561	4606	4647	4684	4717	2863
80	4161	4249	4328	4399	4461	2429
90	3801	3929	4042	4142	4229	2000
100	3484	3645	3786	3911	4020	1580
110	3206	3394	3559	3704	3831	1182
120	2964	3174	3357	3518	3659	839
130	2756	2981	3178	3352	3504	594

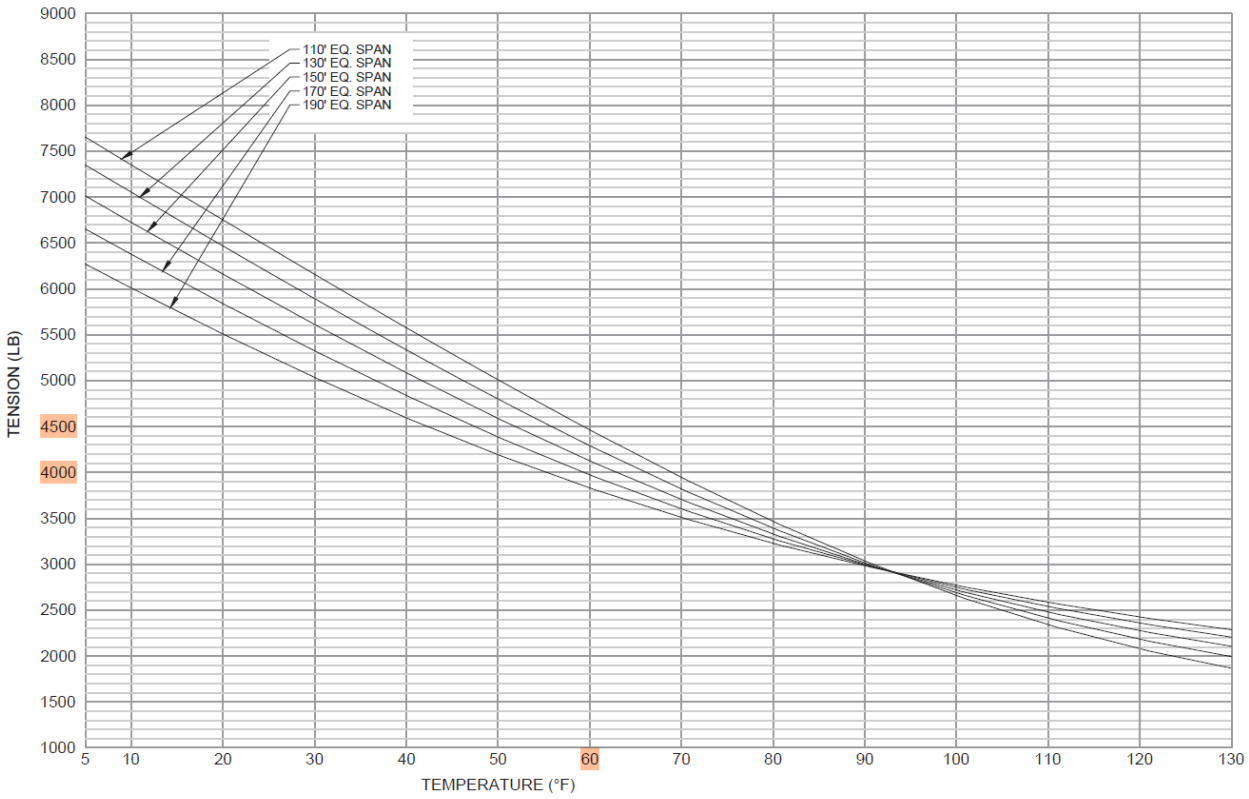


F200 Federal Way LE Messenger Wire (MW) Temp/Tensions Chart (Unloaded) (AT)

UNLOADED MW STRINGING TENSIONS (LB)

TEMP (°F)	EQUIVALENT SPAN (FT)				
	110	130	150	170	190
5	7652	7350	7012	6646	6264
10	7349	7054	6725	6371	6005
20	6749	6469	6160	5834	5503
30	6157	5896	5613	5320	5031
40	5576	5339	5088	4835	4593
50	5010	4804	4592	4385	4192
60	4465	4297	4131	3974	3832
70	3949	3828	3712	3606	3514
80	3472	3403	3339	3283	3235
90	3045	3029	3015	3003	2993
100	2674	2708	2737	2762	2783
110	2362	2438	2502	2556	2602
120	2106	2213	2303	2379	2444
130	1897	2026	2135	2228	2307

UNLOADED MW STRINGING TENSIONS (LB)



10.9 Appendix I – U830 University Link MW and CW Temperature/Tensions Charts

U830 University Link Messenger Wire (CW) Temp/Tensions Chart (Fixed Termination)

U830 Contract IFC Dwg No. L10-JO702, Sheet 547 for CW

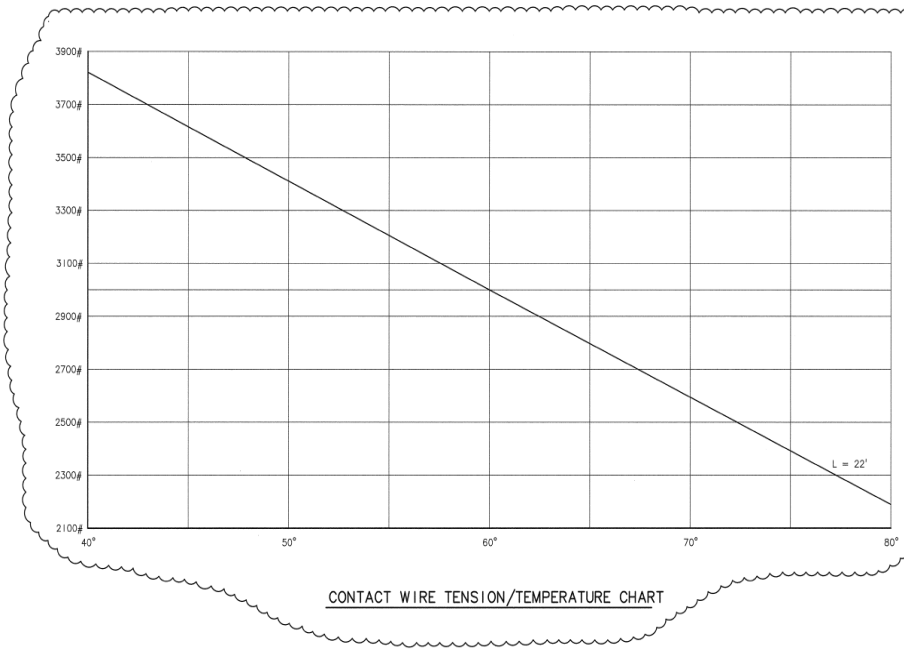
NOTES:

1. TEMPERATURE RANGES INSIDE THE TUNNEL ARE 40°F TO 80°F WITH 60°F A MEDIAN TEMPERATURE.
2. EQUIVALENT SPAN IS DETERMINED BY THE FOLLOWING FORMULA:

$$\sqrt{\frac{L_1^3 + L_2^3 + L_3^3 + \text{---} + L_n^3}{L_1 + L_2 + L_3 + \text{---} + L_n}}$$

WHERE L_1, L_2, L_3 ---L_n ARE THE LENGTHS OF THE SPANS IN EACH TENSION LENGTH (SPRING OR FIXED).

3. FOR INTERMEDIATE EQUIVALENT SPANS BETWEEN THOSE INDICATED IN THE GRAPH, THE TENSION VALUES CAN BE INTERPOLATED ON A STRAIGHT LINE BASIS.



CONTACT WIRE TENSION/TEMPERATURE CHART

	TENSION/ TEMPERATURE CONTACT WIRE:		
	UN–REGULATED TENSION (FIXED SYSTEM)		
L = EQUIVALENT SPAN (FT)	31'	22'	16.5'
TEMPERATURE 40°F	3821#	3825#	3828#
50°F	3410#	3412#	3414#
60°F	3000#	3000#	3000#
70°F	2593#	2589#	2587#
80°F	2189#	2180#	2176#

A

U830 University Link Messenger Wire (MW) Temp/Tensions Chart (Fixed Termination)

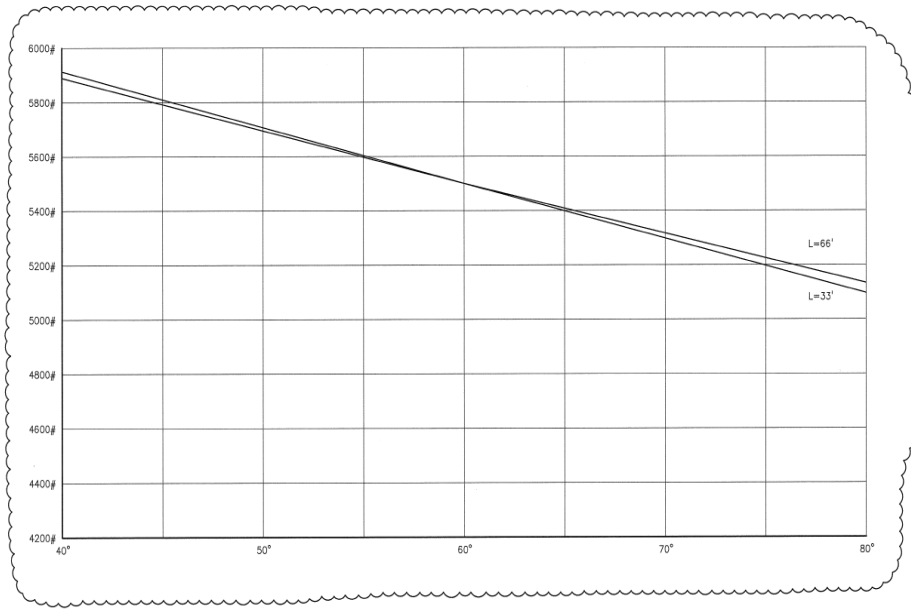
U830 Contract IFC Dwg No. L10-JO703, Sheet 548 for MW

1. TEMPERATURE RANGES INSIDE THE TUNNEL ARE 40°F TO 80°F WITH 60°F A MEDIAN TEMPERATURE.
2. EQUIVALENT SPAN IS DETERMINED BY THE FOLLOWING FORMULA:

$$\sqrt{\frac{L_1^2 + L_2^2 + L_3^2 + \text{---} + L_n^2}{L_1 + L_2 + L_3 + \text{---} + L_n}}$$

WHERE L_1, L_2, L^3 ---L^n ARE THE LENGTHS OF THE SPANS IN EACH TENSION LENGTH (SPRING OR FIXED).

3. FOR INTERMEDIATE EQUIVALENT SPANS BETWEEN THOSE INDICATED IN THE GRAPH, THE TENSION VALUES CAN BE INTERPOLATED ON A STRAIGHT LINE BASIS.



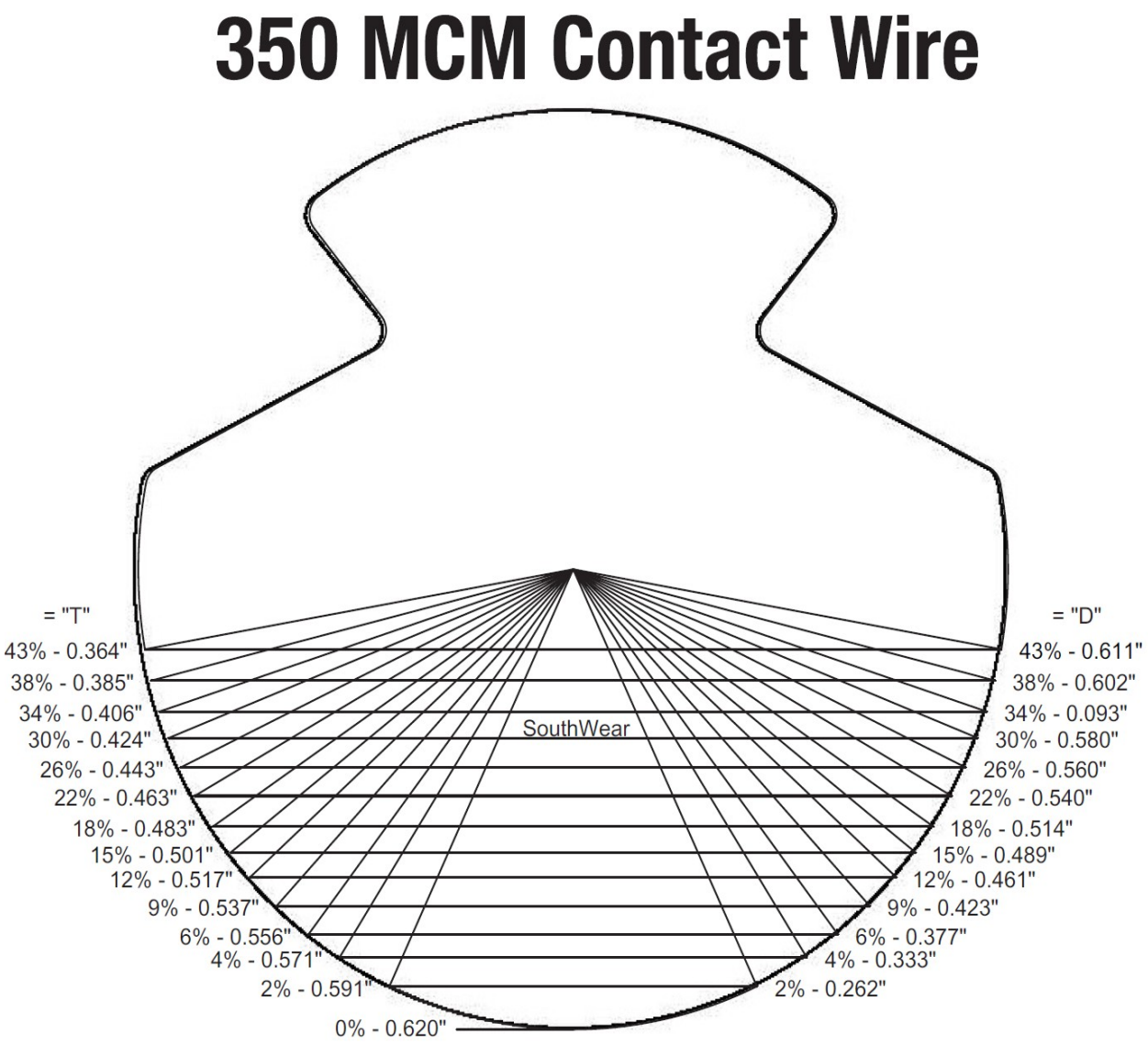
	TENSION/ TEMPERATURE MESSENGER WIRE:	
	UN-REGULATED TENSION (FIXED SYSTEM)	
L = EQUIVALENT SPAN (FT)	66'	33'
TEMPERATURE 40°F	6667#	6733#
TEMPERATURE 50°F	6078#	6115#
TEMPERATURE 60°F	5500#	5500#
TEMPERATURE 70°F	4939#	4890#
TEMPERATURE 80°F	4401#	4289#

A

MESSENGER WIRE TENSION/TEMPERATURE CHART

10.10 Appendix J – Figure 1 – 350 MCM Contact Wire Wear from SouthWear

10.10 Appendix J - Figure Appendices. 350 MCM Contact Wire Wear (detail from SouthWear)



"T" Measurement Top to Bottom on Left **(Measure with caliper in inches)**

0% wear - 0.620" (new wire)

18% wear – 0.483" height to top.

26% wear – 0.483" height to top.

"D" Width of CW contact surface on Right

2% wear – 0.262" contact wire surface width.

18% wear – 0.514" contact wire surface width.

26% wear – 0.560" contact wire surface width.

30% wear – 0.424” height to top. 30% wear – 0.580" contact wire surface width.

Note: Replace CW at 30%.

10.11 Appendix K – Figure 2 – Tensorex C+ Spiral Spring Tensioning Device (by Mosdorfer) for Auto-Tensioned (AT) OCS.

Commissioning Setting example.

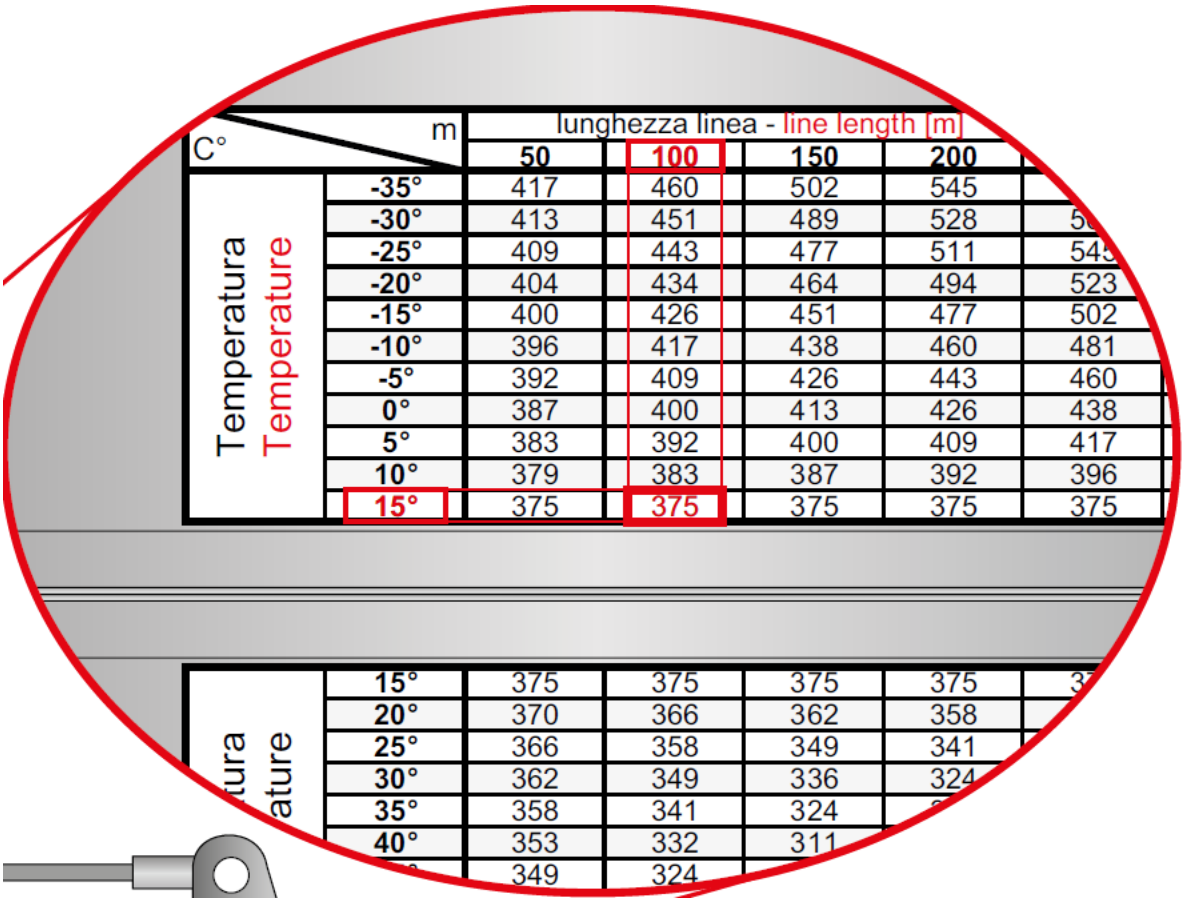
Calculate setting number by crossing 1. the ambient temperature and 2. the tension length on the chart on device.

Ambient Temperature: 15 Celsius (59 Fahrenheit)

OCS wire run tension length (Half-tension length to Midpoint Anchor): 100 meters (328 feet)

Note: This is half the wire run length. From Tensorex C+ to midpoint anchor point.

10.11 Appendix F - Figure 2. Tensorex C+ Commissioning Setting Example



(See Commissioning Setting instructions from manufacturer next page)

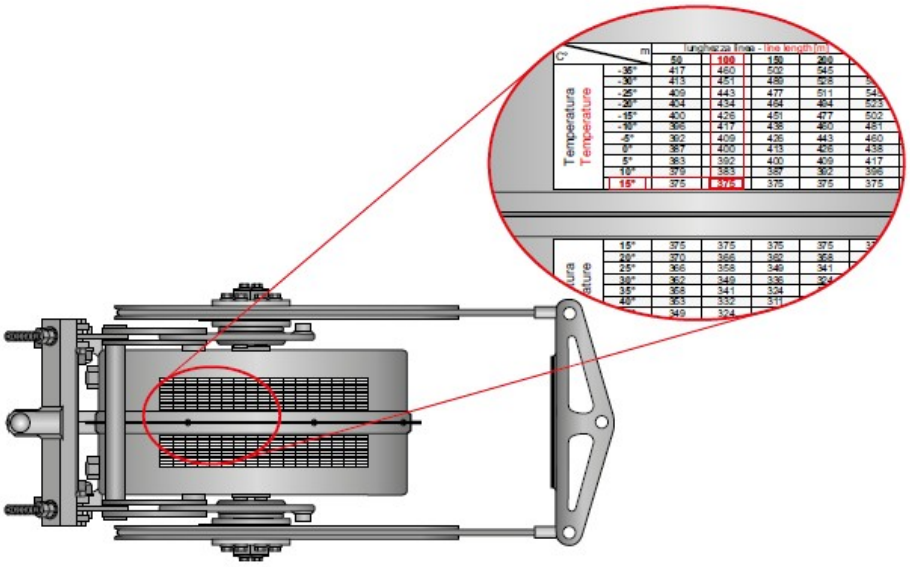
9 Commissioning

9.1 Setting

To commission TENSOREX C+ the current ambient temperature and the tension length are required. Different types of TENSOREX C+ could have different setting tables and setting scales.
The relevant table with the required values sticks on top of TENSOREX C+. And it is also available from MOSDORFER.

1 Calculate the setting number by crossing the ambient temperature and the tension length. (For tension length see drawing in chapter 7.1, page 13.)

The drawing below is just an **example** to simplify the understanding of how to read the table.
Use the table on top of TENSOREX C+.



In this example, the tension length is **100 m** and the ambient temperature **15°C**. Hence, the setting number is **375**.

In order to avoid to go up to TENSOREX C+ during maintenance later on, a copy of the same temperature table should be given to the maintenance team.